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Resource allocation to cell envelopes and the scaling of bacterial growth rate

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Abstract

Although various empirical studies have reported a positive correlation between the specific growth rate and cell size across bacteria, it is currently unclear what causes this relationship. We conjecture that such scaling occurs because smaller cells have a larger surface-to-volume ratio and thus have to allocate a greater fraction of the total resources to the production of the cell envelope, leaving fewer resources for other biosynthetic processes. To test this theory, we developed a coarse-grained model of bacterial physiology composed of the proteome that converts nutrients into biomass, with the cell envelope acting as a resource sink. Assuming resources are partitioned to maximize the growth rate, the model predicts that the growth rate and ribosomal mass fraction scale negatively, while the mass fraction of envelope-producing enzymes scales positively with surface-to-volume. These relationships are compatible with growth measurements and quantitative proteomics data reported in the literature.

1. Introduction

The rate of cell growth varies across bacterial species. The inhabitant of salt marshes, *Vibrio natriegens*, divides in 10 min [1], whereas the causal agent of gum disease, *Treponema denticola*, takes 20 h to divide [2]. Given the centrality of the growth rate in physiology, ecology, and evolution, efforts have been invested in understanding the causes of this variation. Previous meta-analyses have reported that the overall pattern of variation is predictable: bacterial species with larger cell volume tend to grow faster. This observation was made in both heterotrophic [3–5] and autotrophic bacteria [6, 7]. The causes of this scaling are unclear. Eukaryotic growth rates scale negatively with body size, implying that the theories explaining this pattern—such as those invoking transport-related constraints [8, 9] and limits imposed by self-shading of chlorophyll [10]—are fundamentally inadequate in accounting for the opposite scaling observed in bacteria.

Here, we examine an alternative explanation based on the trade-off in the allocation of resources between fast growth and envelope production.

The resource in our model is the building block (e.g. amino acids, the compounds of the carbon metabolism, etc), which is invested in either the proteome or the cell envelope, but not both of these traits at the same time. Hence, investing more resources in one trait comes at the expense of the other, and the organism has to find a way to optimally allocate resources such that it balances demand for both. Namely, funneling resources toward the proteome—which maximizes the growth rate because the proteome makes itself—has to be balanced with investments into the cell envelope, ensuring enough surface features to cover the underlying cytoplasm. The balance shifts with the size of the bacterial species. As the cell size becomes smaller, a larger fraction of the resources are diverted to the envelope, and fewer resources are left for ribosomes, which, in turn, means that the cell grows more slowly. It was initially suggested that this constraint might impose the limit on the smallest size that a cell can attain [11]. This idea has been further used to explain the lower limit on the size of the photoautotrophic organism and why phytoplankton growth rates appear to increase with cell volume [12, 13]. Our goal is to formalize

this theory and investigate whether it can explain the growth scaling in heterotrophic bacteria. We start by imagining the fastest-growing cell as a bag of self-replicating ribosomes, following the previously developed framework [14, 15]. However, the growth rate is depressed below this perfect state because the cell has to invest resources in (1) machinery that acquires and converts nutrients to fuel ribosomes, and (2) cell envelope, which is necessary for the cell to maintain the proper shape.

We formalize the afore-mentioned verbal argument in sections 2.1 and 2.2, and then use this framework to obtain the simple analytical solution for maximal attainable growth rate given a particular cell size (section 2.3). These predictions are tested using quantitative proteomic data of the model bacterium, *Escherichia coli* (section 3.1). The cross-species scaling expectations are compared against data on bacterial growth rates, cell sizes, and proteome compositions in sections 3.2 and 3.3. The implications of the model for the scaling of the metabolic rate is outlined in section 3.4. We end by discussing the competing explanations in section 4.

2. Materials and methods

2.1. Derivation of the steady-state growth rate

In our model, a cell is composed of two metabolite species and three protein species (figure 1). The external nutrient concentration is assumed to be constant, thus mimicking the nutrient-replete conditions when bacteria are grown in the lab. These nutrients are taken up and converted into building block b which corresponds to amino acids. Although we refer to species l as lipids, this group includes all molecules used in the construction of the cell envelope, such as peptides and saccharides (including components of membrane lipoproteins and peptidoglycan). We use upper-case symbols to refer to absolute abundances, and lower-case symbols for relative abundances or concentrations. All chemical reactions obey Michaelis–Menten kinetics, and we assume that half-saturation constants (K_M) for all reactions are identical. We ultimately focus on the special case when all reactions are operating at saturation ($K_M = 0$) because this permits a simple analytical solution for the steady-state of various cellular features, and we only vary K_M to compare this limiting behavior to a more general case when cellular enzymes are not saturated. Table 1 outlines the meaning of symbols that are used throughout the main text.

Our model is similar to those proposed in [16], with two important differences. First, the cell divides once it reaches a critical volume, rather than a critical abundance of a particular protein controlling cell division. We use this model because it was the easiest to deal with analytically, and other cell

division mechanisms (such as the ‘adder’ principle which seems to hold in multiple bacterial species and according to which adds a constant volume between birth and division [17]) are unlikely to affect the overall scaling patterns, as the envelope imposes a burden regardless of the exact molecular details underlying the cell division. Second, we introduced the degradation of macromolecules, and the production of an additional metabolite, which is the envelope component. Perhaps most importantly, we obtained a simple analytical solution to the maximal growth rate in the limit of saturation, whereas the model in [16] is explored only numerically. This allows one to quickly derive relationships between proteomic mass fractions and the growth rate under various perturbations, and use these to infer model parameters. The model is also similar to the one reported in [18], as these authors also include protein degradation, but differs in that it posits second-order mass action kinetics, whereas we let reactions run according to Michaelis–Menten kinetics. Lastly, we show that their analytical solution for maximal growth rate reduces to our solution, in the limit of no envelope burden and enzyme saturation.

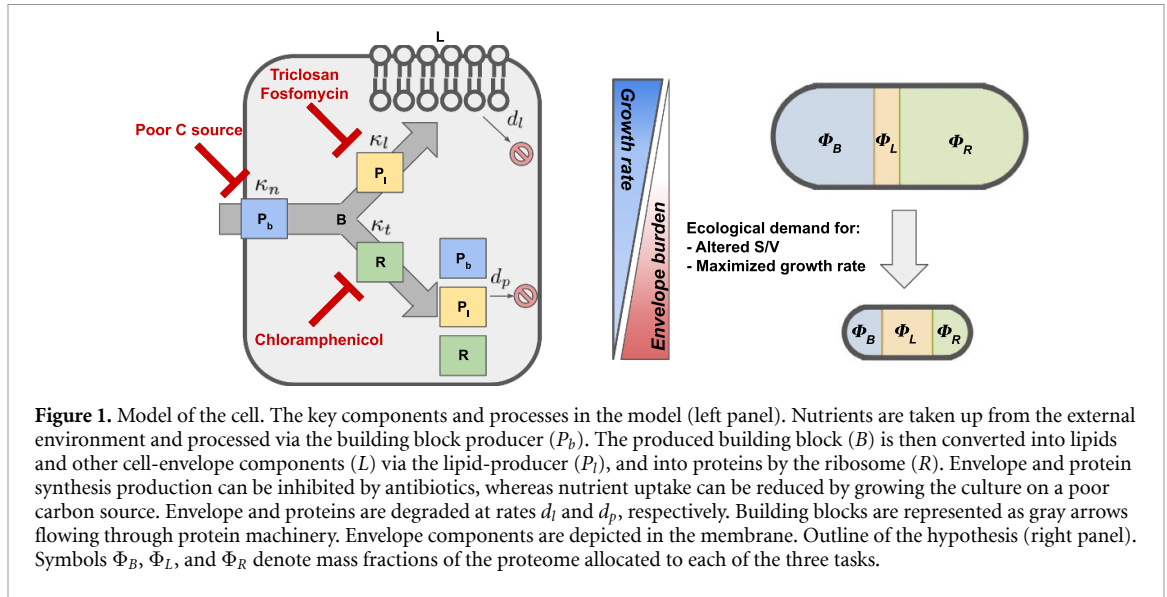
The time-evolution of metabolite concentrations are:

$$\dot{b} = k_n p_b(t) - (k_t r(t) + k_l p_l(t)) \frac{b(t)}{K_M + b(t)} - \lambda(t) b(t) \quad (1a)$$

$$\dot{l} = \frac{k_l}{m_s} p_l(t) \frac{b(t)}{K_M + b(t)} - (d_l + \lambda(t)) l(t). \quad (1b)$$

The production of b occurs at rate k_n , which represents a pseudo-first-order rate constant that depends on the nutrient status of the environment inhabited by the cell. Rate constants k_n , k_l , and k_t are the turnover numbers for reactions of nutrient processing, envelope synthesis, and translation, while m_s is the size an envelope unit expressed in terms of the number of building blocks. Envelope components are eliminated from the cell by degradation at rate d_l , and by dilution due to growth at rate $\lambda(t)$. Note that the growth rate is time-dependent because it is a function of state variables (i.e. molecular abundances). We neglect the degradation of the free amino acids and focus only on the turnover of cell-envelope components.

Protein species p_b , p_l , and r are produced by ribosomes which represent the autocatalytic part of the cell. All proteins have associated degradation rates d_p , with the exception of ribosomes which are reported to be remarkably stable in exponentially growing cells [19]. Sizes of the metabolic protein (P_b and P_l) and the ribosome are m_p and m_r , respectively. The time-evolution equations are identical in form to equations for metabolites:



$$\dot{p}_b = \frac{k_t}{m_p} r(t) \phi_b \frac{b(t)}{K_M + b(t)} - (d_p + \lambda(t)) p_b(t) \quad (2a)$$

$$\dot{p}_l = \frac{k_t}{m_p} r(t) \phi_l \frac{b(t)}{K_M + b(t)} - (d_p + \lambda(t)) p_l(t) \quad (2b)$$

$$\dot{r} = \frac{k_t}{m_r} r(t) \phi_r \frac{b(t)}{K_M + b(t)} - \lambda(t) r(t) \quad (2c)$$

with the only notable difference that—given the finite ribosomal pool—ribosomes have to be partitioned between different protein species, and this is denoted with ϕ_x , which represents the fraction of total ribosomal concentration that is allocated to translation of protein species x (x being either r , b , or l). We will assume that each molecular species contributes to volume proportional to its size in amino acids. For example, ribosomes are about $20\times$ larger than other metabolic proteins, contributing $20\times$ more to the cell volume. Concentrations of each species are defined as the abundance (labeled with the capital letter) divided by cytoplasmic volume V :

$$x(t) = \frac{X(t)}{V(t)}, \quad x \in \{b, l, p_b, p_l, r\}, \quad X \in \{B, L, P_b, P_l, R\} \quad (3a)$$

$$V(t) = m_r R(t) + m_p P_b(t) + m_p P_l(t) + B(t). \quad (3b)$$

The volume V corresponds to the internal volume of the cell where the chemical reactions take place, and including the envelope in the volume of the cell would imply that one can slow down chemical reactions by, say, increasing the thickness of the cell wall. As this is a non-sensical conclusion, we assume that L does not contribute to cytoplasmic volume. The cell is assumed to grow exponentially, such that $\dot{V} = \lambda(t)V(t)$. Given that both volume and concentrations are time-dependent, the left hand side of equations (1a)–(2c) will take the form:

$$\dot{x} = \frac{\dot{X}}{V(t)} - \frac{X(t)}{V(t)} \frac{\dot{V}}{V(t)} = \frac{\dot{X}}{V(t)} - x(t) \lambda(t). \quad (4a)$$

Substituting equation (4a) in equations (1a)–(2c) and rearranging yields:

$$\dot{B} = k_n P_b(t) - (k_l R(t) + k_l P_l(t)) \frac{B(t)/V(t)}{K_M + B(t)/V(t)} \quad (5a)$$

$$\dot{L} = \frac{k_l}{m_s} P_l(t) \frac{B(t)/V(t)}{K_M + B(t)/V(t)} - d_l L(t) \quad (5b)$$

$$\dot{p}_b = \frac{k_t}{m_p} R(t) \phi_b \frac{B(t)/V(t)}{K_M + B(t)/V(t)} - d_p p_b(t) \quad (5c)$$

$$\dot{p}_l = \frac{k_t}{m_p} R(t) \phi_l \frac{B(t)/V(t)}{K_M + B(t)/V(t)} - d_p p_l(t) \quad (5d)$$

$$\dot{R} = \frac{k_t}{m_r} R(t) \phi_r \frac{B(t)/V(t)}{K_M + B(t)/V(t)}. \quad (5e)$$

When cells are in a steady-state, all cellular components grow exponentially at a constant rate $\tilde{\lambda}$ which is the steady-state growth rate. We use this property to find the steady-state of the dynamical system. More precisely, we have:

$$X(t) = X(0) e^{\tilde{\lambda} t} \implies \dot{X} = \tilde{\lambda} X(0) e^{\tilde{\lambda} t}. \quad (6a)$$

Note that $X(0) = k\tilde{X}$, where $\tilde{X}$ is abundance at cell division time when the cell is in the steady-state, and k is $1/2$ if the abundance doubles over the life cycle. Substituting equation (6a) in the system (5a)–(5e) leads to:

$$\tilde{\lambda} \tilde{B} = k_n \tilde{P}_b - (k_l \tilde{R} + k_l \tilde{P}_l) \frac{\tilde{B}/\tilde{V}}{K_M + \tilde{B}/\tilde{V}} \quad (7a)$$

$$\tilde{\lambda} \tilde{L} = \frac{k_l}{m_s} \tilde{P}_l \frac{\tilde{B}/\tilde{V}}{K_M + \tilde{B}/\tilde{V}} - d_l \tilde{L} \quad (7b)$$

$$\tilde{\lambda} \tilde{P}_b = \frac{k_t}{m_p} \tilde{R} \phi_b \frac{\tilde{B}/\tilde{V}}{K_M + \tilde{B}/\tilde{V}} - d_p \tilde{P}_b \quad (7c)$$

Table 1. List of symbols used in the text and their meaning.

Symbol	Meaning
$p_b(t)$	The relative abundance of building block-producing enzyme at time t
$p_l(t)$	The relative abundance of envelope-producing enzyme at time t
$r(t)$	The relative abundance of ribosomes at time t
$l(t)$	The relative abundance of envelope component
$b(t)$	The relative abundance of building blocks
$P_b(t)$	The absolute abundance of building block producer
$P_l(t)$	The absolute abundance of envelope producer
$R(t)$	The absolute abundance of ribosomes
$L(t)$	The absolute abundance of envelope component
$B(t)$	The absolute abundance of building blocks
ϕ_b	The ribosome fraction allocated to translation of building block producers
ϕ_l	The ribosome fraction allocated to translation of envelope producers
ϕ_r	The ribosome fraction allocated to translation of ribosomes
Φ_B	The proteome mass fraction allocated to translation of building block producers
Φ_L	The proteome mass fraction allocated to translation of envelope producers
Φ_R	The proteome mass fraction allocated to translation of ribosomes
m_p	Length of building block producers and envelope producers in amino acids
m_r	Length of ribosome in amino acids
m_s	Length of the unit of envelope (lipids, peptides, and sugars) in amino acids
k_n	The turnover number (k_{cat}) of building block producers
k_l	The turnover number (k_{cat}) of envelope producers
k_t	The turnover number (k_{cat}) of ribosomes
d_p	Protein degradation rate
d_l	Envelope degradation rate
K_M	Michaelis–Menten constant for all chemical reactions
Π	Surface area-to-cytoplasmic volume ratio
β	The number of envelope units per unit of surface area
γ	The number of amino acids in proteome per unit of cytoplasmic volume
ϵ	The resource cost of the unit S/V ($m_s\beta/\gamma$)
$V(t)$	Volume of the cytoplasm
$\lambda(t)$	The specific growth rate at time t

$$\tilde{\lambda}\tilde{P}_l = \frac{k_t}{m_p}\tilde{R}\phi_l\frac{\tilde{B}/\tilde{V}}{K_M + \tilde{B}/\tilde{V}} - d_p\tilde{P}_l \quad (7d)$$

$$\tilde{\lambda}\tilde{R} = \frac{k_t}{m_r}\tilde{R}\phi_r\frac{\tilde{B}/\tilde{V}}{K_M + \tilde{B}/\tilde{V}}. \quad (7e)$$

An intuitive interpretation of these equations is that the rate at which molecules are synthesized and degraded ultimately equals the rate at which they are diluted by growth. Note that $\tilde{V}$ is the steady-state cell volume which is by definition the critical volume at which the cell divides. To simplify downstream

expressions, let $\kappa_n = k_n/m_p$, $\kappa_t = k_t/m_r$, $\kappa_l = k_l/m_p$. These are rate constants scaled by the size of a particular enzyme, capturing how fast an enzyme is relative to its size. Solving for molecular abundances yields:

$$\tilde{B} = \left(\frac{\kappa_n\phi_b - \tilde{\lambda} - d_p}{\tilde{\lambda}} - \frac{\kappa_l\phi_l}{\kappa_t\phi_r} \right) \chi \quad (8a)$$

$$\tilde{L} = \frac{\tilde{\lambda}\phi_l\kappa_l/m_s}{\kappa_t\phi_r(d_l + \tilde{\lambda})} \chi \quad (8b)$$

$$\tilde{P}_b = \frac{\phi_b}{m_p} \chi \quad (8c)$$

$$\tilde{P}_l = \frac{\phi_l}{m_p} \chi \quad (8d)$$

$$\tilde{R} = \frac{(\tilde{\lambda} + d_p)\phi_r}{m_r\tilde{\lambda}} \chi \quad (8e)$$

where

$$\chi = \frac{\tilde{\lambda}^2 \kappa_t \phi_r K_M \tilde{V}}{(\tilde{\lambda} - \kappa_t \phi_r) (\kappa_l \phi_l \tilde{\lambda} + \kappa_t (\tilde{\lambda} + d_p - \kappa_n \phi_b) \phi_r)}.$$

By definition, proteome mass fractions are:

$$\begin{aligned} \Phi_B &= \frac{m_p \tilde{P}_b}{m_p \tilde{P}_b + m_p \tilde{P}_l + m_r \tilde{R}}, \\ \Phi_L &= \frac{m_p \tilde{P}_l}{m_p \tilde{P}_b + m_p \tilde{P}_l + m_r \tilde{R}}, \\ \Phi_R &= \frac{m_r \tilde{R}}{m_p \tilde{P}_b + m_p \tilde{P}_l + m_r \tilde{R}}. \end{aligned} \quad (9)$$

By replacing steady-state abundances in equation (9) with solutions in equations (8a)–(8e), we have:

$$\begin{aligned} \Phi_B &= \frac{\tilde{\lambda}\phi_b}{\tilde{\lambda} + d_p\phi_r}, \quad \Phi_L = \frac{\tilde{\lambda}\phi_l}{\tilde{\lambda} + d_p\phi_r}, \\ \Phi_R &= \frac{(d_p + \tilde{\lambda})\phi_r}{\tilde{\lambda} + d_p\phi_r}. \end{aligned} \quad (10)$$

In the limit when there is no protein degradation ($d_p = 0$), the proteomic mass fractions are equal to the fraction of ribosomes allocated to the synthesis of a particular protein component. This result was originally obtained in [15]. However, we are still lacking the solution for the steady-state growth rate $\tilde{\lambda}$.

Given that volume increases exponentially and from the definition of the cell volume (equation (3b)), we have:

$$\lambda(t) = \dot{V}/V(t) = m_p \left[\kappa_n \frac{P_b(t)}{V(t)} - \left(\kappa_l \frac{B(t)/V(t)}{K_M + B(t)/V(t)} \frac{P_l(t)}{V(t)} + d_p \left(\frac{P_b(t)}{V(t)} + \frac{P_l(t)}{V(t)} \right) \right) \right] \quad (11a)$$

$$\tilde{\lambda} = m_p \left[\underbrace{\kappa_n \frac{\tilde{P}_b}{\tilde{V}}}_{\text{aa influx}} - \left(\underbrace{\kappa_l \frac{\tilde{B}/\tilde{V}}{K_M + \tilde{B}/\tilde{V}} \frac{\tilde{P}_l}{\tilde{V}}}_{\text{outflux to } L} + \underbrace{d_p \left(\frac{\tilde{P}_b}{\tilde{V}} + \frac{\tilde{P}_l}{\tilde{V}} \right)}_{\text{dissipation via degradation}} \right) \right] \quad (11b)$$

where equation (11b) has been obtained by applying (6a) to molecular abundances in (11a). The steady-state growth rate is equal to the rate at which the building blocks are generated from the acquired nutrients minus the building blocks that are diverted into cell envelope synthesis and thus do not contribute to cytoplasmic volume production or are dissipated through protein degradation occurring at a rate d_p . Note that diversion to lipids and other envelope constituents does not explicitly enter the growth rate because L does not contribute to the cytoplasmic volume. However, lipid production does affect the growth rate by altering the amount of available building blocks $\tilde{B}$, and by diverting proteome allocation to P_l . Finally, substituting equations (8a)–(8e) into (11b), we retrieve a cubic polynomial:

$$a_0 + a_1 \tilde{\lambda} + a_2 \tilde{\lambda}^2 + a_3 \tilde{\lambda}^3 = 0 \quad (12)$$

with the following coefficients:

$$a_0 = 0 \quad (13a)$$

$$a_1 = \kappa_t^2 \phi_r^2 (d_p - \kappa_n \phi_b) \quad (13b)$$

$$a_2 = \kappa_t \phi_r ((1 + K_M) \kappa_n \phi_b + \kappa_l \phi_l - d_p (1 + K_M (\phi_b + \phi_l)) + \kappa_t \phi_r) \quad (13c)$$

$$a_3 = -(1 + K_M) \kappa_l \phi_l - \kappa_t \phi_r \quad (13d)$$

equation (12) has two non-zero roots:

$$\tilde{\lambda} = \frac{-a_2 \pm \sqrt{a_2^2 - 4a_3a_1}}{2a_3}. \quad (14)$$

In the case of the positive branch, $\tilde{\lambda} = 0$ when $\phi_r = 0$ and $\tilde{\lambda} = \kappa_t$ when $\phi_r = 1$. This is clearly incorrect, given that $\tilde{\lambda}$ should be zero both when the cell does not have ribosomes (because there is nothing to build the cell) and when the cell contains only ribosomes without any other protein components (because there is nothing to deliver building blocks to the ribosomes). On the contrary, the negative branch attains values of zero both when $\phi_r = 0$ and when $\phi_r = 1$, so we take this solution as biologically meaningful. Finally, we obtain the steady-state abundances by substituting equation (14) in equations (8a)–(8e). The resulting analytical solutions match the numerically-integrated equations (figure 2).

Although the explicit solution for the maximal $\tilde{\lambda}$ is prohibitively difficult to obtain, we can gain some insight by examining boundary cases when the cell does not have a membrane ($\phi_l = 0$) and cellular processes are infinitely fast. When $\kappa_n \rightarrow \infty$, $\tilde{\lambda} \rightarrow \kappa_t/(1 + K_M)$, and, conversely, as $\kappa_t \rightarrow \infty$ we have $\tilde{\lambda} \rightarrow \kappa_n - d_p$. These two asymptotic results occur because the cell that instantaneously converts nutrients to building blocks will saturate the downstream translation machinery and the growth rate will be determined by the rate at which ribosomes operate. Conversely, when ribosomes are infinitely fast, the growth rate is set by how fast the building blocks are supplied to protein synthesis machinery.

2.2. Optimization problem of the cell

It is a working hypothesis that the steady-state growth rate $\tilde{\lambda}$ is a function of the ribosome partitioning parameters ϕ , and the cell has a task to find an optimal partitioning across different proteome components such that $\tilde{\lambda}$ is maximized. This maximization is assumed to be achieved by homeostatic mechanisms that constantly take input from the environment and adjust the proteome composition accordingly. Note, however, that the cell could attain maximal growth even if the partitioning parameters were hard-coded in the genome and thus cannot be actively adjusted (i.e. when homeostatic mechanisms are absent). In that case, the process of finding the peak is governed by natural selection and other evolutionary forces. Hence, our model holds regardless of whether the actual cells have regulatory mechanisms or not.

Growth rate maximization proceeds with two constraints. First, the fractions of ribosomes allocated to the production of three different proteome sectors have to sum to unity (c1). Second, the surface-to-volume ratio has to satisfy the constraint based on the shape of the cell (c2):

$$\begin{aligned} &\text{Maximize } \tilde{\lambda} \\ &\text{subject to: } \phi_b + \phi_l + \phi_r = 1 \end{aligned} \quad (c1)$$

$$\frac{\tilde{L}/\beta}{\tilde{V}/\gamma} = \Pi. \quad (c2)$$

Note that V is the cytoplasmic volume, and thus the Π is the ratio of cell surface area to cytoplasmic volume. Parameters γ and β are unit conversion

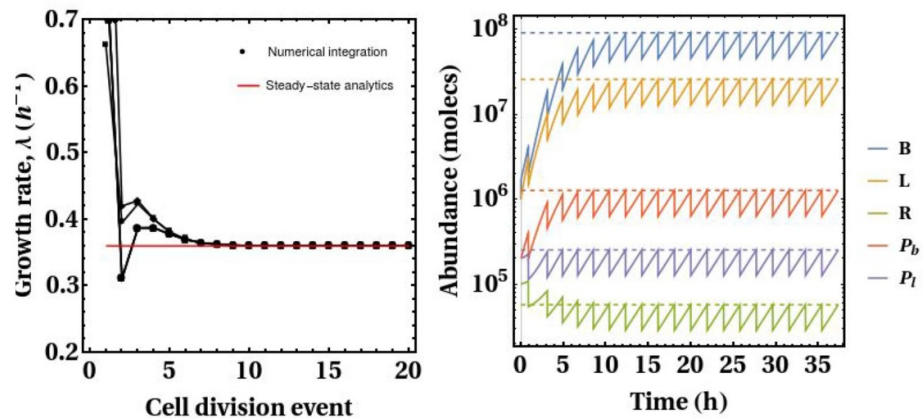


Figure 2. Comparison of the analytical and numerical solution for the cell in the steady-state. The left panel shows a close match of the steady-state growth rate (solid red line) relative to numerically integrated system of ODEs (black dots). Each set of points corresponds to a different set of initial molecular abundances. On the right, a similar match between analytics and numerics also exists for abundances of five molecular species; each gray dashed line represents the steady-state analytical solution given by equations (8a)–(8e). Parameters: $\kappa_l = 50 \text{ h}^{-1}$, $\kappa_r = 1 \text{ h}^{-1}$, $d_p = d_l = 0.1 \text{ h}^{-1}$, $K_M = 0.01$, $V = 10^9 \text{ aa}$, $\phi_b = 0.5$, $\phi_l = 0.1$, $\phi_r = 0.4$.

factors corresponding to the number of amino acids per unit of cell volume (molecules μm^{-3}), and the number of lipids/envelope components per unit of cell surface area (molecules μm^{-2}). These two parameters are treated as species-specific and independent of the growth conditions. While *E. coli* growth rate experiences five-fold variation across media with different carbon sources, the cytoplasmic density in *E. coli* is roughly constant [20]. With respect to β , our rationale was that if envelopes are built out of the similar macromolecules (phospholipids, glycolipids, peptidoglycan disaccharides, and so on) the surface area of envelope constituents should be roughly constant, and thus the number of them that the cell can fit per unit of surface area of the envelope should be independent of the cell size or the growth rate. To intuitively understand the optimization problem, consider a landscape of ribosomal partitioning parameters and the resulting growth rates (figure 3). In the absence of geometric constraint (c2), the cell maximizes growth rate by (1) completely abolishing expression of the envelope-producing enzyme (when $\phi_l = 0$), and (2) by optimally expressing ribosomes (when $0 < \phi_r < 1$); this is because envelope-producing enzyme acts as a burden that diverts resources from other proteome components, and because the cell has to balance the production of proteins (for which high ribosomal expression is required) and the production of building blocks that fuels the translation (for which low ribosomal expression is required). These two aspects explain why the growth rate is a monotonically decreasing function of ϕ_b , and a non-monotonic function of ϕ_r .

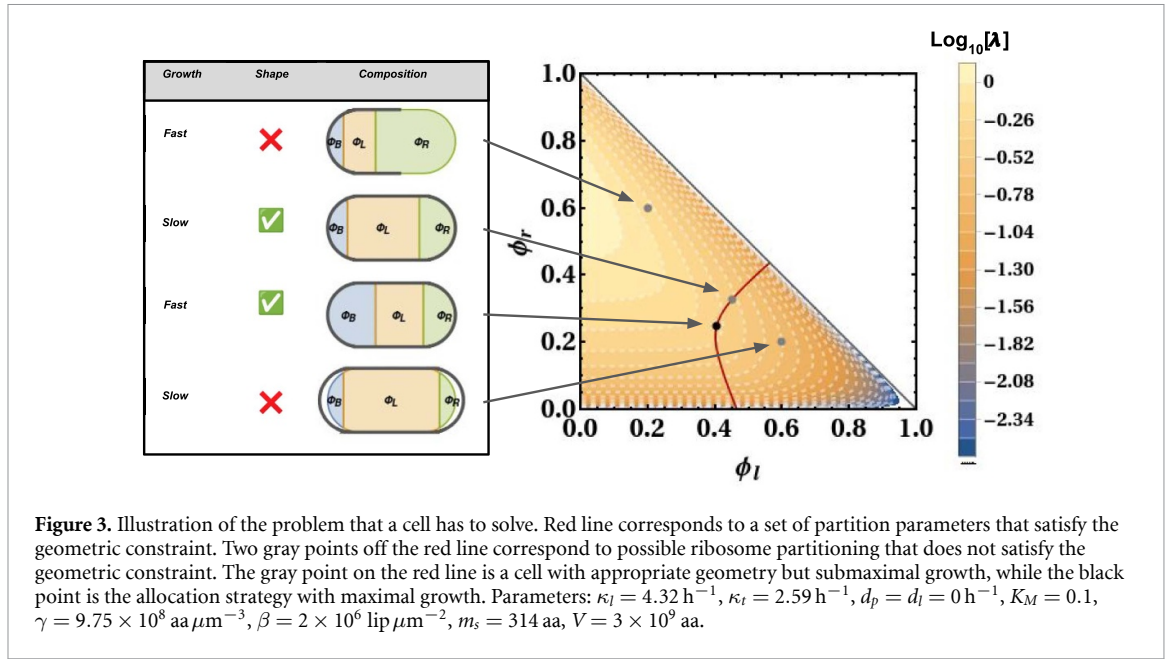
Now suppose that the cell maximizes the growth rate, while at the same time having to maintain a surface-to-volume ratio dictated by the cell's geometry; out of all partitioning parameters, only a subset satisfies this constraint and these parameters fall

onto the red line in figure 3. For example, a cell that has a low expression of the envelope-producing enzyme (the left-most gray point in the landscape) will have a high growth rate but will not have enough envelope to cover its cytoplasm, thus making it a non-viable option. On the other hand, a cell that has a highly expressed envelope producer (the right-most gray point) will make too much membrane relative to its volume, thus causing the cell to wrinkle. However, a cell with intermediate values of ϕ_l will produce just enough envelope to cover its cytoplasm (the intermediate gray point), and the maximal growth rate is achieved by the adjustment of partitioning parameters along this line (black point in the landscape).

The developed model assumes that the S/V is constant across the cell cycle, which is wrong, given that the cell has to change the shape and size as it grows. Unfortunately, we currently cannot derive a more general model which accounts accurately for changes in shape as the individual cell evolves. However, the variation in S/V across the cell cycle is much smaller than the cross-species range of S/V that we focus on (section S1.4), so it should not strongly affect the reported derivations.

2.3. Explicit solution for maximal growth under saturation kinetics

Although obtaining the analytical solution for maximal steady-state growth rate is prohibitively difficult, one can obtain this property for a special case when all enzymes are saturated, such that chemical reactions obey first-order mass-action kinetics. Indeed, most *E. coli* enzymes have K_M lower than their substrate's concentrations [21]. This reduces a nonlinear system of equations (7a)–(7e) to a system of linear equations that can be readily solved for the partition parameters ϕ :



$$\tilde{\lambda}\tilde{L} = \frac{k_l}{m_s}\tilde{P}_l - d_l\tilde{L} \quad (15a)$$

$$\tilde{\lambda}\tilde{P}_b = \frac{k_t}{m_p}\tilde{R}\phi_b - d_p\tilde{P}_b \quad (15b)$$

$$\tilde{\lambda}\tilde{P}_l = \frac{k_t}{m_p}\tilde{R}\phi_l - d_p\tilde{P}_l \quad (15c)$$

$$\tilde{\lambda}\tilde{R} = \frac{k_t}{m_r}\tilde{R}\phi_r. \quad (15d)$$

The steady-state growth is then obtained from equation (11b) by setting $K_M = 0$:

$$\tilde{\lambda} = m_p \left(\kappa_n \frac{\tilde{P}_b}{\tilde{V}} - \kappa_l \frac{\tilde{P}_l}{\tilde{V}} - d_p \left(\frac{\tilde{P}_b}{\tilde{V}} + \frac{\tilde{P}_l}{\tilde{V}} \right) \right). \quad (16a)$$

Substituting equation (16a) in equations (15a)–(15d):

$$\tilde{L} = \frac{\kappa_l \phi_l \kappa_t \phi_r}{m_s (\kappa_n \phi_b - \kappa_l \phi_l - d_p (\phi_b + \phi_l)) (\kappa_t \phi_r + d_l)} \tilde{V} \quad (17a)$$

$$\tilde{P}_b = \frac{\kappa_t \phi_r \phi_b}{m_r (\kappa_n - d_p) \phi_b + m_r (\kappa_l + d_p) \phi_l} \tilde{V} \quad (17b)$$

$$\tilde{P}_l = \frac{\kappa_t \phi_r \phi_l}{m_r (\kappa_n - d_p) \phi_b + m_r (\kappa_l + d_p) \phi_l} \tilde{V} \quad (17c)$$

$$\tilde{R} = \frac{\phi_r (\kappa_t \phi_r + d_p)}{m_r (\kappa_n - d_p) \phi_b + m_r (\kappa_l + d_p) \phi_l} \tilde{V}. \quad (17d)$$

By substituting abundances in equation (9) for equations (17b)–(17d) it immediately follows that the mass fractions of protein sectors are:

$$\Phi_B = \frac{\kappa_t \phi_b}{\kappa_t + d_p}, \quad \Phi_L = \frac{\kappa_t \phi_l}{\kappa_t + d_p}, \quad \Phi_R = \frac{\kappa_t \phi_r + d_p}{\kappa_t + d_p}. \quad (18)$$

Note that substitution of equations (17a)–(17d) in equation (16a) yields $\tilde{\lambda} = \kappa_t \phi_r$, meaning that the

growth rate is simply proportional to the fraction of ribosomes that are allocated to ribosome translation. The cell, however, cannot allocate the entirety of its ribosomes to this task because this would halt the production of other necessary protein components. Therefore, to find the optimal partitioning of ribosomes that maximizes the growth rate, one has to impose additional algebraic constraints:

$$m_p \kappa_n \tilde{P}_b = m_p \kappa_l \tilde{P}_l + m_r \kappa_t \tilde{R} \quad (19a)$$

$$\frac{\tilde{L}/\beta}{\tilde{V}/\gamma} = \Pi. \quad (19b)$$

The first algebraic constraint (equation (19a)) reflects the fact that maximal growth is achieved when the building block influx matches the outflux. If influx is higher than the outflux, then building blocks unnecessarily accumulate in the cell, and if the outflux is higher, then the building block pool will become completely depleted and the chemical reactions will halt. Because B is being produced and consumed at matching rates, the building block pool does not grow over time, and we retrieve the constraint by setting the left-hand side of equation (7a) to zero and $K_M = 0$. The second expression (equation (19b)) imposes a constraint on the amount of resources that has to be diverted into cell envelope to ensure a proper cell shape.

To obtain the optimal partitioning parameters ϕ that satisfy these constraints, we substitute for $\tilde{P}_b$, $\tilde{P}_l$, and $\tilde{R}$ with the solutions (17a)–(17d). While there are three ϕ parameters, there are only two degrees of freedom given that $\phi_b = 1 - \phi_r - \phi_l$. Thus, we have a system of two linear equations in two unknowns, which we solve to obtain the optimal partitioning of the proteome such that the flux of resources is balanced between catabolic and anabolic processes, and the cell has a proper shape given its size:

$$\phi_r^{\text{opt}} = \frac{\kappa_l \kappa_t (\kappa_n - d_p) - d_l (\kappa_l + \kappa_n) (\kappa_t + d_p) \epsilon \Pi}{\kappa_l \kappa_t (\kappa_n + \kappa_t) + \kappa_t (\kappa_l + \kappa_n) (\kappa_t + d_p) \epsilon \Pi} \quad (20a)$$

$$\phi_l^{\text{opt}} = \frac{(\kappa_t + d_p) (\kappa_n d_l + \kappa_t (\kappa_n + d_l - d_p)) \epsilon \Pi}{\kappa_l \kappa_t (\kappa_n + \kappa_t) + \kappa_t (\kappa_l + \kappa_n) (\kappa_t + d_p) \epsilon \Pi} \quad (20b)$$

Substituting ϕ_r^{opt} and ϕ_l^{opt} in equations (17a)–(17d), one retrieves molecular abundances in terms of model parameters. Substitution of the same into equation (18), allows one to obtain expressions for optimal proteome mass fractions. Finally, substitution in formula for $\tilde{\lambda}$ (equation (16a)) yields a maximal steady-state growth rate $\tilde{\lambda}_{\text{max}}$:

$$\tilde{\lambda}_{\text{max}} = \frac{\kappa_t (\kappa_n - d_p)}{\kappa_n + \kappa_t} \frac{1}{1 + \Theta}, \quad (21a)$$

$$\Theta = \frac{(\kappa_n + \kappa_l) (\kappa_t + d_p) (\kappa_n d_l + (\kappa_n + d_l - d_p) \kappa_t) \epsilon \Pi}{(\kappa_n + \kappa_t) (\kappa_l \kappa_t (\kappa_n - d_p) - d_l (\kappa_n + \kappa_l) (\kappa_t + d_p) \epsilon \Pi)}, \quad (21b)$$

where $\epsilon = m_s \beta / \gamma$ is the resource cost of the unit S/V , or the resource investment in a unit of surface area per unit of cytoplasmic volume. For example, if $\epsilon = 2$, then each added unit of the surface area requires twice as many amino acids relative to the added unit of cytoplasmic volume.

Intuitively, the first term in equation (21a) is the steady-state growth rate in the limit of no additional resource sink, and the second term represents a deviation from maximal achievable growth due to cell envelope production. The parameter Θ is the bioenergetic cost of cell envelope production, which depends not only on the actual amount of resources that go into this cellular feature ($\epsilon \Pi$) but also on the rates of all cellular processes. Previous theoretical developments represent a special case of equation (21a). In the limit of no investment into cell envelope ($\Pi = 0$), equation (21a) reduce to those reported in [18], and taking this further to the case without protein degradation ($d_p = 0$) yields the result in [15].

While the explicit solution for Θ appears complicated, some insight can be gained by looking at the special case when there is no degradation ($d_l = d_p = 0$):

$$\Theta = \frac{\frac{\kappa_l + \kappa_n}{\kappa_l} \epsilon \Pi}{\frac{\kappa_n + \kappa_t}{\kappa_t}} \quad (22)$$

The term $\epsilon \Pi$ corresponds to the resource cost of producing the entire envelope structure. Parameter ϵ can be interpreted as the envelope cost of the cell with $\Pi = 1 \mu\text{m}^{-1}$. This is because γ is the total number of amino acids per unit volume and $m_s \beta$ is the total number of amino acid equivalents required to build the unit of the cell surface. Given that Π is the surface-to-cytoplasmic volume ratio, the whole term $\epsilon \Pi$ is the cost of producing surface relative to the whole amino acid budget of the cytoplasmic volume. The fractional term in equation (22) captures the cost of producing the enzyme machinery that builds the actual envelope. One cellular process supplies the building blocks at the per unit proteome rate κ_n , and

two other cellular processes are competing for this common pool at the per unit proteome rates of κ_l and κ_t . For instance, if κ_l is decreased, the cell maximizes growth rate by overexpressing cell envelope-producing machinery in order to compensate for the low per unit proteome rate, thus increasing the total costs of the envelope.

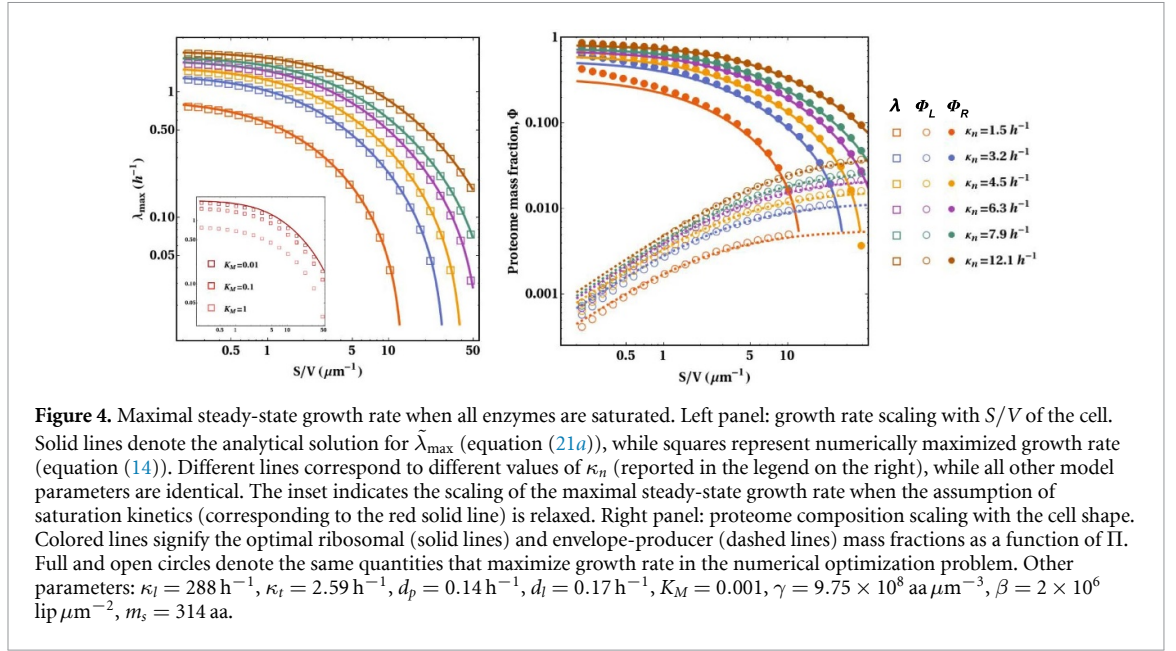
The expressions for proteomic mass fractions are retrieved by substituting partition parameters in equation (18) for the optimal partition that maximizes the growth rate (equations (20a) and (20b)):

$$\Phi_R = \frac{\kappa_n - d_p}{\kappa_n + \kappa_t} \frac{1}{1 + \Theta} \quad (23)$$

$$\Phi_L = \frac{\kappa_n - d_p}{\kappa_n + \kappa_t} \frac{\Theta}{1 + \Theta}. \quad (24)$$

One sees that ribosomal mass fraction is a monotonically decreasing and the envelope-producer mass fraction is a monotonically increasing function of the surface-to-volume ratio, Π .

The analytical solution for $\tilde{\lambda}_{\text{max}}$, and optimal Φ_R and Φ_L in the saturation limit was cross-validated by comparison to the numerically maximized $\tilde{\lambda}$ with algebraic constraints of the finite ribosomal pool (c1) and geometric constraint on the cell shape (c2), as described in section 2.2, and optimization was performed using the Nelder–Mead algorithm. We generally find an excellent correspondence between analytics and numerics for both the growth rate and proteome composition (right panel in figure 4). When enzymes in the model operate far from the saturation limit (i.e. when $K_M \gg 0$), the analytics break down (see inset). Note that large K_M has a similar effect on scaling as a reduction in nutrient quality of the media. Therefore, although the analytical solution neglects the presence of the aqueous phase—metabolites and associated water molecules—inside the cell, the incorporation of this property affects the intercept but not the overall scaling pattern. For



a fuller analysis of growth-impeding effects of cell envelope, see section S1.1.

2.4. Parameterization of the model

Equation (21a) has six parameters (κ_n , κ_l , κ_t , ϵ , d_l , d_p) and one variable (Π). Our goal was to constrain the rates of chemical reactions to the values occurring in *E. coli*, a species that is well characterized at physiological level, and then ask how the growth rate would scale if the variable Π was altered. That is, we are interested in understanding how the growth rate would scale if all bacterial species were biochemically identical to an *E. coli* cell, and only differed in cell size.

We estimated the resource cost of the envelope in a cell with a unit S/V , ϵ , by assuming that resource investment is equal to the mass M of a structure. Given that our model consists of the whole envelope (lipids, peptides, and sugar attachments) with mass M_{env} and proteins of mass M_{prot} , and that envelope and proteins account for roughly 30% and 55% of total cell mass M_T (see [22], and chapter 2 in [23]), and $\Pi = 5$ (table 2):

$$\begin{aligned} \epsilon \Pi &= \frac{m_s \beta S}{\gamma V} = \frac{M_{\text{env}}}{M_{\text{prot}}} = \frac{M_{\text{env}}/M_T}{M_{\text{prot}}/M_T} \\ &= \frac{0.30}{0.55} \implies \epsilon = 0.11 \mu\text{m}. \end{aligned} \quad (25)$$

This is a crude estimate because it does not include the direct costs of envelope production and proteome components (i.e. resources needed to convert one chemical compound into another one), but these usually account for a small fraction of the total costs [24], and small changes in ϵ do not affect our conclusion. Similar values are obtained if one calculates ϵ from the ratio of the envelope to protein volumes. Former is $0.14 \mu\text{m}^3$ (see

BNID:104 738 in [23]), while latter occupies $\sim 20\%$ of total cell's volume which is roughly $1 \mu\text{m}^3$.

The degradation rates are estimated separately from the literature. We assumed that the ribosomal degradation rate is zero, and we justify this assertion by three observations. First, ribosomal rRNA is remarkably stable in the exponential and stationary phases, and the degradation occurs only when the culture transitions between these two growth stages [19, 25]. Second, almost all ribosomal proteins have degradation rates close to zero [26]. Third, ribosomal proteins are mutually exchangeable when damaged, meaning that replacement is favored over degradation and re-synthesis [27]. The protein degradation rate was set to 0.05 per hour [28]. We also tried out estimates for various protein classes [29] but found little variation in the scaling pattern (figure S1.8). The envelope degradation rate d_l is difficult to estimate exactly due to the diverse components that make up this structure, but was set to 1 per hour [30]. The latter reports a hypermetric scaling of the peptidoglycan turnover and the growth rate ($d_l = 0.7 \times (\text{Log}[2]/\tau)^{1.38}$). Assuming that an *E. coli* cell divides in 30 min, we have $d_l \simeq 1$. Unfortunately, this is a very crude estimate based on the peptidoglycan degradation rate in *Bacillus subtilis*, which constitutes a large part of its envelope. Peptidoglycan degradation rates are somewhat lower in *E. coli* [31], but on the other hand, that species has an outer membrane containing polysaccharides, and we do not know how fast this component is degraded.

The rate constants of chemical reactions (κ_t , κ_l , κ_n) were inferred from *E. coli* proteomic data across different growth conditions, leveraging growth laws derived in section S1.2. We ignore the intercept of regressions (equations (S1.11a)–(S1.13b)) and infer parameters strictly from the slopes of

Table 2. Parameterization of the model. Standard errors were obtained using error propagation (see section S1.9). † Surface-to-volume ratio represents the average across different dimensions of *E. coli* in our dataset. * Calculated in the text. All rate parameters are Q_{10} -corrected as described in text. ○ Rate constants estimated from the noted study.

Symbol	Definition	Value	Units	Source
$\kappa_n(\text{M63+gly})$	Nutritional capacity in M63 with glycerol	0.478 ± 0.074	h^{-1}	[15], ○
$\kappa_n(\text{M63+glc})$	Nutritional capacity in M63 with glucose	0.985 ± 0.112	h^{-1}	[15], ○
$\kappa_n(\text{cAA+gly})$	Nutritional capacity in cAA and glycerol	1.352 ± 0.11	h^{-1}	[15], ○
$\kappa_n(\text{cAA+glc})$	Nutritional capacity in cAA and glucose	1.863 ± 0.235	h^{-1}	[15], ○
$\kappa_n(\text{RDM+gly})$	Nutritional capacity in RDM with glycerol	2.322 ± 0.284	h^{-1}	[15], ○
$\kappa_n(\text{RDM+glc})$	Nutritional capacity in RDM with glucose	3.441 ± 0.386	h^{-1}	[15], ○
κ_l	The envelope synthesis rate	22.164 ± 4.07	h^{-1}	[32], ○
κ_t	The protein synthesis rate	2.579 ± 0.175	h^{-1}	[32], ○
ϵ	The cost of cell per unit of S/V	0.11	μm	*
Π	Surface-to-volume ratio	5	μm^{-1}	†
d_p	Protein degradation rate	0.011	h^{-1}	[28]
d_l	Cell envelope degradation rate	0.211	h^{-1}	[30]

these relationships. The slope of the regression of the growth rate on the mass fraction of ribosomal proteins allows one to compute κ_t from equation (S1.11a). Similarly, the value of κ_l can be computed from the slope of the regression of the growth rate on the mass fraction of envelope-producing enzymes and equation (S1.11b).

Because κ_n captures both the intrinsic efficiency of metabolism to convert nutrients into building blocks and the nutrient state of the external environment, there will be one κ_n for every medium that *E. coli* is reared in. Intuitively, one would expect minimal media to have lower κ_n than rich media, as the latter contains better quality nutrients and thus leads to more amino acids being generated per unit time per molecule P_b . The parameter κ_n is calculated from the slopes of ribosomal mass fraction across growth conditions with varying concentrations of translation inhibitor, after plugging in values of κ_l , d_p , and $\epsilon\Pi$ in equation S1.12a. The values are computed for the following media: M63 with glycerol, M63 with glucose, casamino acids with glycerol, casamino acids with glucose, rich defined media with glycerol, and rich defined media with glucose. Since our data for growth rates across different bacterial species are normalized to 20°C using Q_{10} correction with the coefficient of 2.5, we also normalized the inferred rate constants using the same method. Estimates of rate constants for *E. coli* are reported in table 2.

One can intuitively explain the estimated rate constants by using logic outlined in [33]. It takes roughly 10^9 glucose molecules to produce the entire carbon skeleton of an *E. coli* cell. Given that an amino acid has, on average, about the same number of carbon atoms as a single glucose molecule (5.35 C-atoms), one could say that the cell requires 10^9 amino acid-equivalents for its construction. Therefore, with roughly 3×10^6 copies of metabolic proteins in the cell, each building block producer making $k_n = \kappa_n m_p = 2 \times 325$ amino acids per hour, it will take ~ 30 min for the metabolic proteins to generate enough amino acids to replace the cell. This is also

the experimentally measured cell division time of an *E. coli* cell grown under favorable conditions. For the protein synthesis rate (i.e. the elongation rate) κ_t , note that each ribosome converts $\kappa_t \times m_r$ amino acids into proteins (where the length of the ribosome m_r is 7336 amino acids). This means that the inferred translation rate k_t in our model is about 25 amino acids per ribosome per second, which is close (albeit slightly larger) to empirical values for an *E. coli* cell [23]. The length of metabolic protein was taken to be the median length of an *E. coli* protein (325 aa from [34]), and ribosome length is set to the total number of amino acids in ribosomal proteins (7336 aa from [35]).

3. Results

The model of the cell developed in section 2 purports to explain two phenomena: (1) the optimal partitioning of the proteome when the same bacterial species is reared under different growth conditions; and (2) the scaling of the growth rate and proteome composition across bacterial species of different shapes and sizes. Both sets of predictions are tested in the ensuing sections. Applying the developed theory to data on growth and proteome composition of the model organism *E. coli*, we first show that our model yields a good qualitative description of physiological responses to changes in the environment. We then use this correspondence between theory and data to estimate the parameter values that yield a good quantitative fit as well (section 3.1).

By parameterizing our model with values obtained from *E. coli*, one can address how the growth rate is supposed to scale with the shape and size of the cell (section 3.2). This implicitly assumes that all bacterial species are biochemically identical to *E. coli*, such that the rates of all chemical reactions are the same. Lastly, in an attempt to simplify the expectation and remove dependence on parameters that are difficult to estimate, we look at the special case when degradation is absent, and the envelope is

costly enough such that $\tilde{\lambda}_{\max}$ is inversely proportional to Π (section 3.3).

3.1. Proteome reallocation across growth conditions

The internal homeostatic mechanisms allow the cell to allocate the proteome to different cellular tasks such that the growth rate is maximized. While *E. coli* cells achieve this via alarmone ppGpp [36], our model is agnostic of the exact mechanism and simply assumes that such resource-tuning strategy exists. One can express the proteome mass fraction of a particular component as a linear function of growth rate when the latter is altered via changes in growth conditions (derived in section S1.2)). Given that there are three cellular processes in our representation of the cell (building block production, cell envelope synthesis, and protein synthesis), we ask how the proteome composition changes when each process is perturbed. More precisely, we are interested in how the proteome composition changes as the growth rate is modulated by changing the values of κ_n , κ_l , or κ_t .

The first expectation is that the ribosomal mass fraction of the proteome Φ_R scales with the growth rate across different nutrient conditions $\tilde{\lambda}_N$ as (see equation (S1.11a)):

$$\Phi_R = \Phi_R^{\{N,\min\}} + \frac{1}{\kappa_t + d_p} \tilde{\lambda}_N. \quad (26)$$

The cell re-balances building block supply and biosynthesis demand by allocating proteome to the limiting process, which is, in this case, protein synthesis. The proteomic data for *E. coli* qualitatively corroborate this expectation (figure 5, upper left panel). Data from different studies appear to show a small amount of variation, but the overall trend is strong. The second expectation is that the envelope-producer fraction increases with the growth rate under nutrient perturbation (see equation (S1.11b)):

$$\Phi_L = \Phi_L^{\{N,\min\}} + \frac{\epsilon \Pi(\tilde{\lambda}_N)}{\kappa_l} \tilde{\lambda}_N. \quad (27)$$

The slope captures the fact that faster growth implies that a cell has to produce its envelope faster. Given that each envelope-producer operates at a constant rate, the only way this constraint can be met is by allocating more enzymes to this task.

When inferring parameters, one must account for the fact that Π might depend on $\tilde{\lambda}$. Π is largely independent of the growth conditions when the latter is varied with a translation-inhibiting antibiotic (figure S1.6). However, Π decreases with the growth rate when the bacteria are cultured with different carbon sources [37–39]. We account for this confounding factor by plugging the equation for the relationship $\Pi \sim \tilde{\lambda}_N$ derived from empirical studies into equation (27), and then infer κ_l using the non-linear least squares method (see section S1.5 for

details). While the proteomic data on the envelope-producer component is less accurate compared to ribosomal proteins (probably owing to low abundance of envelope-producers), the predicted positive scaling still occurs in the two most comprehensive proteomic studies (pink circles and orange squares in lower left panel of figure 5).

The third expectation is concerned with scaling of ribosomal mass fraction under translation inhibition (see equation (S1.12a)):

$$\Phi_R = \Phi_R^{\{T,\max\}} - \frac{\kappa_l + (\kappa_n + \kappa_l) \epsilon \Pi}{\kappa_l (\kappa_n - d_p)} \tilde{\lambda}_T. \quad (28)$$

As the concentration of a translation-inhibiting drug is increased, so is the fraction of inhibited ribosomes. The cell compensates for this inhibition by overexpressing ribosomes, which causes a negative scaling between Φ_R and $\tilde{\lambda}_T$. We see a good fit to data (upper right panel in figure 5), and similar relationships can be obtained by using the alternative source of data from [39] (figure S1.6).

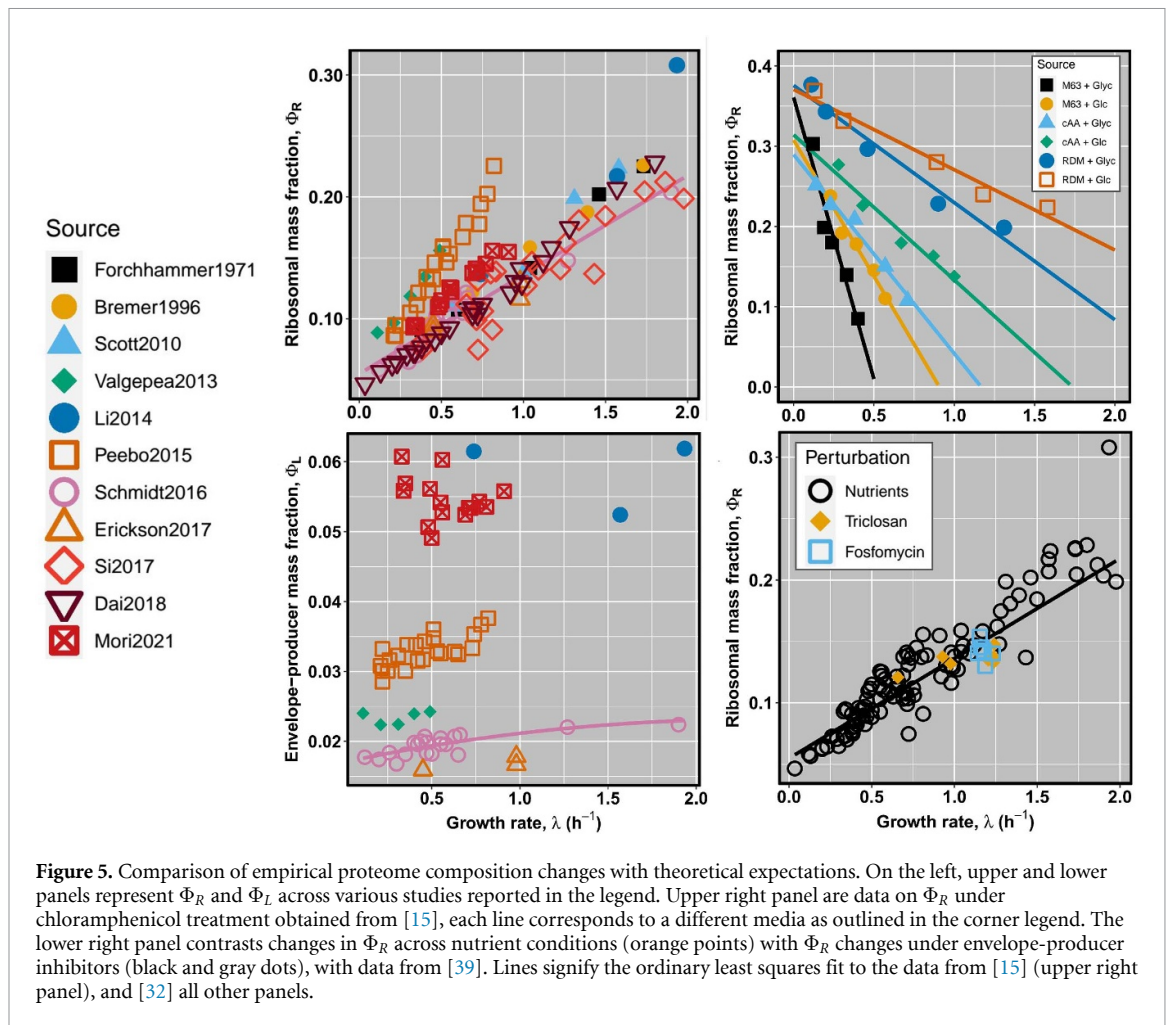
The fourth prediction is that the response of Φ_R to nutrient quality perturbation is the same as the response to envelope-synthesis inhibition (see equation (S1.13a)). This follows from the fact that the relationship

$$\Phi_R = \Phi_R^{\{L,\min\}} + \frac{1}{\kappa_t + d_p} \tilde{\lambda}_L \quad (29)$$

is identical to equation (26). Thus, the prediction is that points from two types of perturbation experiments ought to fall onto the same line. Indeed, the treatment of *E. coli* culture with fosfomycin (peptidoglycan synthesis inhibitor) and triclosan (fatty acid synthesis inhibitor) reduces both the growth rate and the ribosomal mass fraction (lower-right panel in figure 5). One can intuitively understand this behavior by noting that the cell responds to the envelope-synthesis inhibition by overexpressing envelope-producer to compensate for the poisoning of the enzymes, which reduces the resources available for ribosomal proteins, thus causing the reduction in their expression.

3.2. Scaling behavior in a fully specified model

The theory is tested by comparing the expected to the observed scaling relationships using growth, cell shape, and proteomic data across heterotrophic bacteria. Details on data collection and normalization are reported in sections S1.6 and S1.7. This requires specifying model parameters. Because we are dealing with a somewhat metabolically homogeneous set of species (i.e. all are heterotrophs), we assume that all the cellular reaction rates are identical, and the only feature that varies across bacteria is Π . Furthermore, most of the maximal growth rates in our dataset were measured in the media containing yeast extract, so we



assume that κ_n is fixed across species as well. The prerequisite for these assumptions to hold is that the type of metabolism of a bacterial species does not correlate with cell size and shape. That is, the variation in metabolism and external environment might affect dispersion around the scaling but does not affect the scaling itself.

The model is determined by six parameters: rate constants of three types of enzymes (κ_n , κ_b , κ_t), degradation rates (d_p , d_l), and resource cost of the unit S/V (ϵ). The degradation rates of lipids, peptidoglycan, and proteins are set to values determined in pulse-chase experiments obtained from the literature (see section 2.4). We take the energetic cost of the envelope, ϵ , from our earlier calculation (equation (25)). Rate constants are not only related to turnover numbers of respective enzyme classes but also depend on other cellular properties that are not explicitly accounted for in the model, such as the nutrient status of the environment, the number of enzymes belonging to a particular sector, the topology of a particular biochemical pathway, the amount of nucleic acid in the cell, and so on. Thus, by estimating rates from the empirical relationships that we derived

in section 3.1, one can ‘collect’ the effects of the explicitly unaccounted cellular properties into these three parameters.

The protein synthesis rate, κ_t , can be estimated from the slope of equation (26) if the degradation rate d_p is known. The envelope synthesis rate, κ_b , is retrieved from the slope in equation (27), if the envelope cost $\epsilon\Pi$ is known. And finally, the nutrient processing rate κ_n is inferred from slopes of changes in Φ_R as the growth rate is perturbed by the addition of translation-inhibiting drug chloramphenicol (equation (28)), after plugging in all of the previously-estimated parameters. Because κ_n also depends on the nutrient quality, the values were inferred for various media ranging from nutrient-depleted (M63 with glycerol) to nutrient-replete (RDM with glucose). All rate constants were estimated by fitting physiological scaling relationships to *E. coli* proteomic data. We first classified each protein into one of the three sectors based on its function (for details, see section S1.7), and then pooled their individual masses to compute the total mass fractions Φ_R and Φ_L . The surface-to-volume ratio was calculated from the linear dimensions of the cell

(section S1.3). Cytoplasmic volume was calculated by first subtracting $2 \times$ thickness of the bacterial envelope from the linear dimensions of the cell, where the latter was taken to be 30 nm [40].

We find that the curves parameterized with κ_n from minimal (black) to rich media (orange) capture most of the variation in growth rate and proteome composition data (figure 6). Hence, in principle, one could attribute the scatter along the y -axis to differences in efficiencies in which bacteria take up and convert the nutrients into building blocks. Furthermore, Φ_L is expected to have a slightly positive and Φ_R a negative scaling with Π , which is not unambiguously corroborated by the data. A large amount of variation in proteomic data around the expectation might reflect growth and temperature differences which we do not know how to correct for. Also, Φ_L is a low abundance proteome sector as it accounts for no more than a few percent of total proteome mass. These two effects might jointly make the data unreliable.

A large variation in growth rate data across bacteria might also stem from variation in morphology (e.g. motility structures), metabolism (e.g. aerobic or anaerobic capabilities), or environment (e.g. from oceanic sediments to digestive tract of mammals). Ideally, one would want to compare the growth rates across species with the same type of metabolism and the same environment and only differ in shape or size. This way, all of the morphological, metabolic, and environmental factors would be controlled, and the entirety of variation in growth can be attributed to variation in size. We attempted to control for these issues by using data from experimental evolution studies, where cells were kept in approximately exponential phase for a long period of time. In the wild, cell size and shape are determined by a myriad of ecological factors [41]. When these cells are taken out of their natural habitat and propagated in the nutrient-rich medium, the ecological factors dictating the size of the cell are removed. For instance, large cells might be deleterious in natural environments owing to size-selective predation but they do not suffer the same disadvantage when reared in the laboratory setting where predators are absent. Under laboratory selection for fast growth, a mutant with lower S/V has a growth advantage over the wild-type due to lower relative investment in envelope, and will spread and take over the population. In this situation, our theory predicts three outcomes. First, increasing growth rate should be accompanied by a reduction in S/V . Second, the mass fraction of envelope-producers should decrease because the envelope synthesis becomes less of an obstacle with inflation of the cell size. Third, given a diminished need for synthesizing envelope-producers, a larger mass fraction of the proteome can be allocated to the ribosomal proteins.

To test the first prediction, we analyzed the scaling of the growth rate with S/V in Lenski's

long-term evolution experiment in *E. coli* and with a more recent experimental evolution of *Mycoplasma mycoides*. First, we used S/V in combination with the relative fitness data for four LTEE time points spanning 50 000 generations [42]. Given that relative fitness is the ratio of Malthusian parameters of evolved to ancestral line determined from the head-to-head competition (see [43]), we converted this metric to the growth rate by multiplying it with the growth rate of the ancestral strain (see $V_{\max}$ in [44]). However, because the competition experiment lasts for one day, competing strains enter into stationary phase after a few hours, so the competitive advantage of the evolved line might reflect not only differences in the growth rates but also the ability to survive in the stationary phase. Thus, we sought to cross-validate our approach by including data from [45], which reported the growth rates and cell volumes across the LTEE. Cell volumes were converted to S/V using the empirical scaling $S = 2\pi V^{2/3}$ reported to hold across many bacterial species [46]. Second, we collected relative fitness and cell size of wild-type *Mycoplasma mycoides* and *M. mycoides* whose genome has been minimized by removal of non-essential genes [47]. Comparing LTEE data with our analytical model (figure 6), we find that the observed scaling falls very near to the expected scaling for *E. coli* grown in the media with similar composition (M63) as that used in LTEE (Davis broth with glucose). Note that both indirect estimates (red inverted triangles) and direct measurements (red diamond) of the growth rate fall onto the same line. A similar trend is observed in the experimental evolution of *M. mycoides*.

The latter two predictions were tested using data on the changes in the absolute transcript abundance over the course of the long-term experimental evolution in *E. coli* [48]. To convert transcript abundance into protein abundance, we first computed the number of proteins per transcript for each *E. coli* gene using data in [49], and then multiplied transcript abundance of each gene in long-term evolution experiment with the corresponding conversion factor. One can then obtain the mass fraction of a particular protein by multiplying the abundance with its molecular mass and normalizing by the total mass of the proteome. We find that the average ribosomal mass fraction slightly increases (0.210 ± 0.02 vs 0.218 ± 0.01), while the average envelope-producer mass fraction decreases (0.034 ± 0.001 vs 0.028 ± 0.0008) during 50 000 generations of the experiment. This roughly coincides with the predicted scaling (orange and blue lines in figure 6). It is currently unclear why the mass fractions do not adhere to the same line with the growth rate (black line); perhaps this reflects differences between strains, given that Lenski used derivatives of B strain while the expected scalings were plotted using parameters inferred from experiments with BW25113 and a strain derived from MG1655.

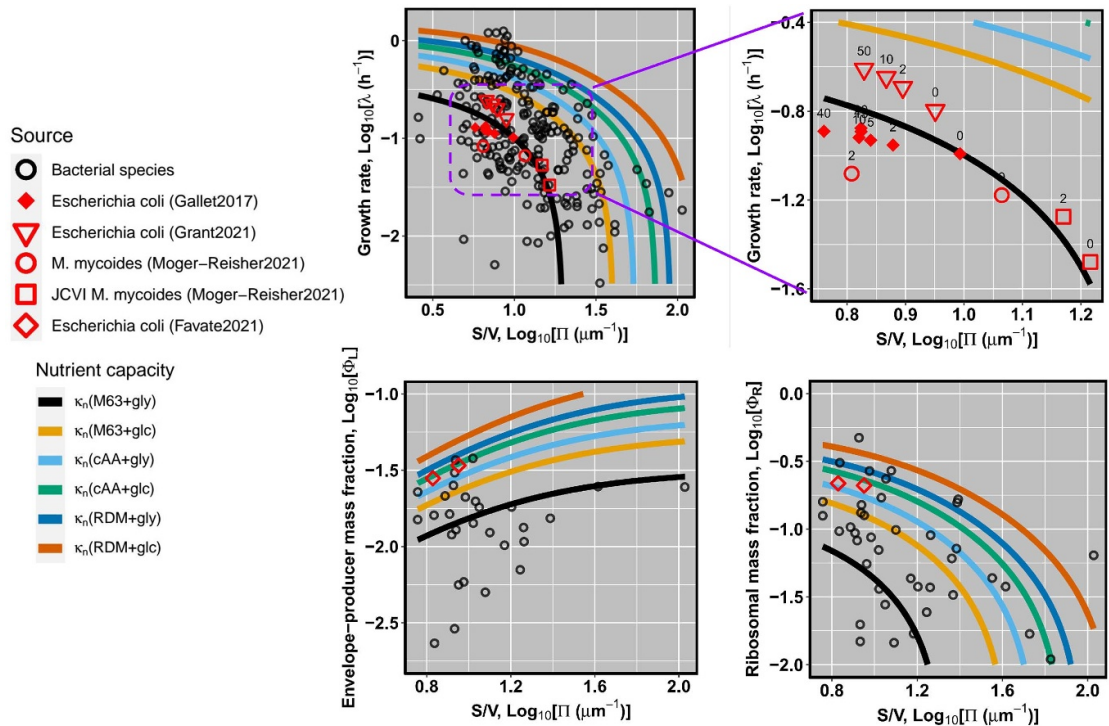


Figure 6. The comparison of expected and observed scaling relationships. Upper row: scaling of growth rate with S/V across bacterial species (left panel), and scaling in long-term evolution experiments (right panel); colored curves denote expected theoretical scaling given by equation (21a) and parameterized with κ_n estimated from various media, as reported in table 2. Values at top points in the upper-right panel indicate the number of generations in thousands since the start of experimental evolution. Lower row: scaling of envelope-producer (left panel) and ribosomal mass fractions (right panel).

3.3. Scaling in the limit of large envelope costs and no degradation

Given the uncertainty of model parameters (especially envelope degradation), we attempted to simplify the theoretical expectation by focusing on the simplest possible case when degradation is absent, and the costs of the envelope are sufficiently high that equation (21a) can be approximated as $\tilde{\lambda}_{\max} \propto \Pi^{-1}$, equation (20a) as $\Phi_R \propto \Pi^{-1}$, and equation (20b) as $\Phi_L \propto \Pi^0$. While this scenario might not be the most realistic, the benefit is that it does not require any further specification of model parameters because the prediction is that the slopes of the regression $\text{Log}_{10}(\tilde{\lambda}_{\max}) \sim \text{Log}_{10}(\Pi)$ and $\text{Log}_{10}(\Phi_R) \sim \text{Log}_{10}(\Pi)$ should be -1 , and the slope of $\text{Log}_{10}(\Phi_L) \sim \text{Log}_{10}(\Pi)$ should be zero. Assuming that the shape is constant across bacteria, similar scalings can be obtained with cell volumes as the independent variable—given that $\Pi \propto V^{-1/3}$. One expects $\text{Log}_{10}(\tilde{\lambda}_{\max}) \sim 0.33 \text{ Log}_{10}(V)$, $\text{Log}_{10}(\Phi_R) \sim 0.33 \text{ Log}_{10}(V)$, and Φ_L retains independence as with S/V . We use these expectations as the null hypothesis for the slopes in the regression analysis. In addition to scaling expectations, we also wish to test whether surface-to-volume is a better predictor of the growth rate than the cell volume. To this end, we performed OLS regression with either S/V or V as independent variables, and the growth rate, or proteomic mass fractions as the dependent variable of various bacterial species (table 3). While

our theory relates the growth rate to the ratio of the surface area to cytoplasmic volume (S/V_{cyt}), we also examined whether the dependent variables scale with surface area to the total cell volume (S/V_{tot}). In total, the analysis included the growth rate data from 229 species, and ribosomal and envelope-producer mass fractions from 41 and 30 species, respectively.

Four insights emerge. First, S/V_{cyt} is a moderate predictor of growth rate (adjusted $R^2 = 0.25$), while cell volume appears neither to match the expected 0.33 scaling nor to show any correlation with the growth rate. Second, the observed scaling of growth with S/V is not significantly different from -1 . Third, the scaling of proteomic mass fractions with S/V largely meets the expected slopes of -1 , for Φ_R , and 0 for Φ_L . Fourth, both S/V_{cyt} and S/V_{tot} perform equally well, with S/V_{cyt} being a slightly better predictor of growth than S/V_{tot} (adjusted R^2 of 0.25 vs 0.23) and the scaling exponent being within one standard error of the estimated -1 in the case of the former. Graphic representation of regression analysis is reported in section S1.11.

We cross-validated these results by analyzing a recently-published study [50], which contains data on linear dimensions of the cell and the minimal doubling time. Although we do not find any correlation between cell volume and the growth rate, we find a very weak negative relationship (adjusted $R^2 = 0.08$) between S/V and the growth rate (figure S1.12) with

Table 3. Observed and expected scaling exponent of the growth and proteome composition. Dependent variables are denoted in the first row, while independent variables are listed in the first column. Each table cell contains the observed slope with standard error, while the expected scaling exponent is reported in the parentheses. Regressions that are not significantly different from the expected slope are indicated with * (i.e. where $p > 0.05$). Sample size: $n(\tilde{\lambda}_{\max}) = 229$, $n(\Phi_R) = 41$, $n(\Phi_L) = 30$.

	$\tilde{\lambda}_{\max}$	Φ_R	Φ_L
S/V_{cyt}	$-1.014 \pm 0.116 (-1) *$	$-0.67 \pm 0.20 (-1) *$	$0.186 \pm 0.247 (0) *$
S/V_{tot}	$-1.249 \pm 0.149 (-1) *$	$-0.948 \pm 0.26 (-1) *$	$0.129 \pm 0.37 (0) *$
V_{cyt}	$0.067 \pm 0.06 (0.33)$	$0.189 \pm 0.108 (0.33) *$	$-0.04 \pm 0.139 (0) *$
V_{tot}	$0.019 \pm 0.062 (0.33)$	$0.186 \pm 0.117 (0.33) *$	$0 \pm 0.151 (0) *$

a scaling exponent of -0.97 which is not significantly different from -1 ($p = 0.92$). Given that bacterial species with large S/V (i.e. small coccal or thin helical cells) are frequently parasitic, it is possible that the pattern is mainly driven by large- S/V species growing more slowly because of their natural habitat and not because of trade-offs in investments. If that is the case, then excluding parasitic species from the dataset would eliminate negative scaling. To control for this confounding factor, we separated data into free-living and host-associated species and then performed the regression analysis (see section S1.12). We find a very weak negative relationship between S/V and the growth rate in both of these datasets and pooled data, indicating that preponderance of small-celled species in pathogens is unlikely to account for the scaling pattern.

Thus far, we have assumed that all species have an envelope that is ~ 30 nm thick, like that of an *E. coli* [40]. However, larger bacterial species may have thicker envelopes that require higher resource investments. If resource investments stay the same across the size range, then the variation in growth cannot be explained by the invariant S/V . To account for this possibility, we collected data on envelope thicknesses across 45 species and used these values to compute Π (see section S1.10). We find that $\tilde{\lambda}_{\max}$ scales

with Π with exponent -1.15 (figure S1.9) which is not significantly different from -1 ($p = 0.418$). Similarly, $\tilde{\lambda}_{\max}$ scales with V_{cyt} with exponent 0.37 (table S1.6) which is not significantly different from 0.33 ($p = 0.745$). Therefore, the inclusion of envelope thickness data does not alter the conclusions reached by using fixed thickness.

Three conclusions are reached from the preceding analysis. First, Π is a moderate predictor of $\tilde{\lambda}_{\max}$, and it accounts for roughly a quarter of growth rate variation in the whole dataset. Second, S/V_{cyt} is slightly better predictor of growth rate than S/V_{tot} . Third, the proteome composition qualitatively—but not quantitatively—fits the expected pattern. The ribosomal mass fraction of the proteome scales negatively with Π , albeit with a slope that is shallower than the expected -1 , and the envelope producer mass fraction appears independent of Π , which is in accordance with theory.

3.4. Scaling of the metabolic rate with S/V

While the central trait in the preceding analysis is the growth rate, one can also obtain the expression for the metabolic rate, and the scaling of this feature can be further used to test our theory. One can distinguish between the specific and the total metabolic rate. The former, $\tilde{q}$, is the sum of the rates of cellular reactions taking place in a unit of cytoplasmic volume:

$$\tilde{q} = q_{\text{ATP}} \gamma \left(\underbrace{Q_{aa} m_p \kappa_n \frac{\tilde{P}_b}{\tilde{V}}}_{\text{aa production}} + \underbrace{Q_l m_p \frac{\kappa_l \tilde{P}_l}{m_s \tilde{V}}}_{\text{envelope production}} + \underbrace{Q_t m_r \kappa_t \frac{\tilde{R}}{\tilde{V}}}_{\text{translation}} + \underbrace{Q_{dp} d_p \frac{(\tilde{P}_b + \tilde{P}_l)}{\tilde{V}}}_{\text{protein deg.}} + \underbrace{Q_{dl} d_l \frac{\tilde{L}}{\tilde{V}}}_{\text{envelope deg.}} \right). \quad (30)$$

On the other hand, the total metabolic rate, $\tilde{Q}$, is the sum of the rates of all cellular reactions taking place in the whole cytoplasm; hence, $\tilde{Q} = \tilde{q} \tilde{V}$. Given the empirical relationship $S = 2\pi V^{2/3}$ [46], we can write the total metabolic rate in terms of Π :

$$\tilde{Q} = 8\pi^3 \tilde{q} \Pi^{-3}. \quad (31)$$

Parameter Q_x corresponds to the number of ATPs hydrolyzed by the process x , which can be amino

acid production, cell envelope production, translation, protein degradation, and degradation of the cell envelope. Parameter q_{ATP} is the energy of hydrolysis of a single ATP molecule and is introduced so that the specific metabolic rate can be expressed in joules, and γ is the total number of building blocks needed to produce a unit of internal volume. Substituting $\tilde{P}_b$, $\tilde{P}_l$, $\tilde{R}$, and $\tilde{L}$ for optimal abundances yields the steady-state metabolic rate of the cell as a function of the rate constants and other model parameters. While

the full expression is cumbersome (see supplemental *Mathematica* notebook), one gains insight into the metabolic consequences of envelope production by

focusing on a simpler limiting case when there is no degradation of macromolecules ($d_p = 0$ and $d_l = 0$). In that case, equation (30) reduces to:

$$\tilde{q} = q_{\text{ATP}} \gamma \underbrace{\left[Q_{\text{aa}} + Q_t + \left(Q_{\text{aa}} + \frac{Q_l}{m_s} \right) \epsilon \Pi \right]}_{\text{energy demand per unit of internal volume}} \underbrace{\frac{\kappa_n \kappa_t}{\kappa_n + \kappa_t} \frac{1}{1 + \frac{(\kappa_n + \kappa_l)/\kappa_l}{(\kappa_n + \kappa_t)/\kappa_t} \epsilon \Pi}}_{\text{max. growth rate}}. \quad (32)$$

The term in the square brackets corresponds to the total ATP hydrolyzed to produce a building block and its associated amount of envelope: $Q_{\text{aa}} + Q_t$ is the energy required to convert a nutrient into a building block and put it into a protein which makes up the internal volume, while $(Q_{\text{aa}} + Q_l/m_s)\epsilon\Pi$ is the energy for transformation of a nutrient into a unit of cell envelope that covers the internal volume. Put differently, for every $Q_{\text{aa}} + Q_t$ ATPs that go into production of internal volume the cell has to invest an additional $(Q_{\text{aa}} + Q_l/m_s)\epsilon\Pi$ ATPs towards the covering to maintain proper surface-to-volume ratio. The second term in equation (32) is the maximal steady-state growth rate and sets the time scale of metabolism so that, for example, an increase in rate constants increases the rate of the overall metabolism. While equation (32) indicates that the specific metabolic rate can either increase or decrease with Π , the model parameters for *E. coli* indicate the former.

Intuitively, this is because the surface is largely composed of lipids, while the internal volume is mostly made up of proteins. Because lipids have higher caloric content than proteins and thus require more energy to produce, an increase in Π shifts the investment away from cheaper proteins and toward more expensive lipids. This further implies that the total metabolic rate is roughly proportional to internal cell volume (or inversely proportional to the cube of Π), albeit with some deviations from this proportionality due to dependence of $\tilde{q}$ on Π . Large S/V have a larger total metabolic rate than expected from Π^{-3} due to the synthesis of costly envelope components. We now wish to test the scaling predicted by equations (30)–(31). The metabolic rate depends on the parameters reported in table 2, and on additional parameters indicating the amount of ATP each reaction uses. These are estimated in section S1.14, and reported in table S1.10. Following the approach in section 3.2, we compare the theoretical expectation to the specific metabolic rates across bacteria that engage in aerobic oxidation and to the changes in metabolic rate across the long-term evolution experiment in *E. coli* (for details on data collection see section S1.15).

Two key results are worth noting. First, contrary to our expectation, we find that the observed specific metabolic rate decreases with the surface-to-volume

ratio across bacterial species (black points and black dashed line in upper panels of figure 7). The straightforward interpretation of this result is that our theory cannot account for differences in $\tilde{q}$ across Π . However, noting that the negative relationship is weak ($p = 0.014$, adjusted $R^2 = 0.09$), it is possible that such correlation would disappear upon inclusion of additional species with large S/V , such as spirochetes and helical mollicutes. Furthermore, the 10-fold variation of metabolic rate for any value of S/V may make it difficult to statistically detect a mild 1.5-fold increase in $\tilde{q}$ with Π predicted by the theory. On the contrary, $\tilde{q}$ does increase with Π in experimentally evolved lines of *E. coli*, but the increase is larger than expected (red markers and red dashed line). Second, the observed total metabolic rate roughly matches the expected scaling (lower panels of figure 7), with metabolic rates in experimentally evolved lines being elevated relative to cross-species data and having a shallower slope.

4. Discussion

Motivated by the observation that larger bacteria tend to grow faster, we proposed a theory that explains this relationship in terms of a trade-off between investment into surface features and biosynthetic machinery that builds the cell. By formalizing this verbal statement into a quantitative model, we obtained predictions that we sought to test with published data. Four conclusions are reached. First, the model recapitulates the previous physiological responses ($\Phi_R \sim \tilde{\lambda}_N$, and $\Phi_R \sim \tilde{\lambda}_T$). It also points to new ones ($\Phi_L \sim \tilde{\lambda}_N$, and $\Phi_R \sim \tilde{\lambda}_L$), although further experimental work is needed given the variability in the scaling relationships between different studies. Second, it explains why the surface-to-volume ratio is a better predictor of growth rate than the volume itself. Third, the model correctly predicts negative scaling of ribosomal content with S/V . On the other hand, the purported mild positive scaling of Φ_L is not unambiguously corroborated by proteomic data. Fourth, while the presented theory gives a rough account of the scaling of the total metabolic rate with the surface-to-volume ratio, the data do not corroborate the expected increase in the volume-specific

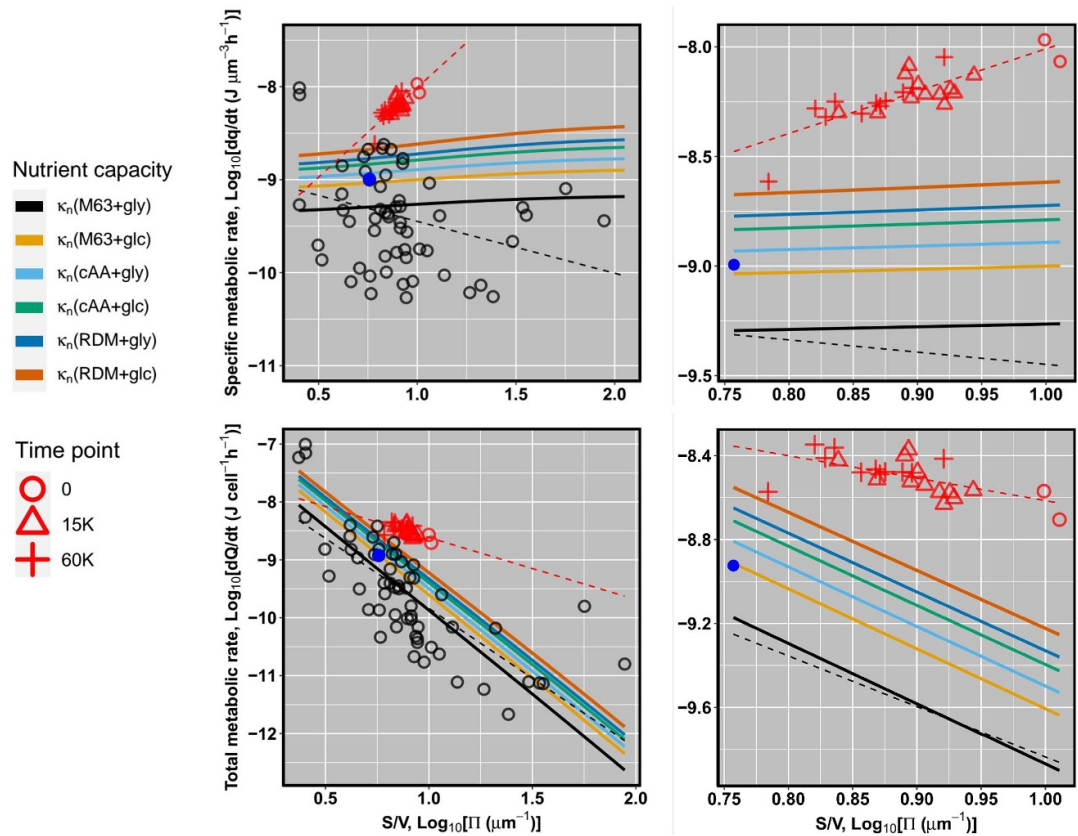


Figure 7. The scaling of the metabolic rate with S/V . Upper panels depict the scaling of total metabolic rate: On the left, comparison of measured metabolic rate and theoretical expectation (equation (30)) for different values of nutritional capacity κ_n inferred earlier. Black points are metabolic rates across bacterial species (most of it from [51]), while red markers indicate metabolic rate measurements reported in [52]. These are labeled according to the number of generations since the start of the experimental evolution. Red dashed line is the regression for long-term evolution experiment: $\log_{10}(dq/dt) = -9.939(\pm 0.255) + 1.93(\pm 0.285) \log_{10} \Pi$ ($p < 10^{-6}$, adjusted $R^2 = 0.65$). Black dashed line is the regression for cross-species data: $\log_{10}(dq/dt) = -8.891(\pm 0.209) - 0.558(\pm 0.22) \log_{10} \Pi$ ($p = 0.014$, adjusted $R^2 = 0.09$). Blue filled circle corresponds to the metabolic rate of *E. coli* in our cross-species dataset and is given for comparison. On the right, same as on the left but focused on *E. coli* data. Lower panels denote the scaling of specific metabolic rate: Scaling of total metabolic rate with S/V (equation (31)). The regression equation in the black dashed line is $\log_{10}(dQ/dt) = -7.422(\pm 0.263) - 2.415(\pm 0.276) \Pi$ ($p < 10^{-11}$, adjusted $R^2 = 0.57$). The regression in red dashed line is $\log_{10}(dQ/dt) = -7.544(\pm 0.255) - 1.07(\pm 0.285) \Pi$ ($p = 0.001$, adjusted $R^2 = 0.35$).

metabolic rate. This could be because our theory lacks an unknown factor determining how cell size affects the metabolic rate, or because the expected increase is too small to statistically detect at the moment given the limited data.

We attempted to control for factors that might affect the scaling. By cross-validating with an independently published data, we find that our conclusions are not an artifact of the dataset. By performing separate regressions with data on host-associated and free-living species, we excluded the possibility that the overall scaling is caused by small-celled species being dominated by organisms selected for slow growth to avoid excessively damaging their host. By including data on envelope thickness when calculating S/V_{cvt} , we excluded the possibility that the envelope becomes thicker with cell size, thus increasing the relative investment into the envelope as cells become larger [53, 54]. By comparing growth rates across experimentally evolved studies where fast growth is selected, we controlled for other factors—such as the

environment that bacteria inhabit—that confound the scaling of the growth rate. Indeed, the scaling identical to cross-species comparisons holds as well.

In addition to explaining the scaling of growth rate across bacterial species, the developed theory may also explain an increase in cell size with an increasing growth rate in experimentally evolved lines. Mechanistically, a genotype with an increased cell size will reduce the metabolic burden of synthesizing the cell envelope and, hence, confer a higher growth rate. The cell size in the long-term experimental evolution in *E. coli* exhibits a step-like trajectory, where relatively long intervals of cell volume constancy are interrupted by short bursts of inflation of cell volume [55]. Interestingly, both relative fitness and cell size experience increase in roughly three bursts in the first 1500 generations that appear to coincide with each other [56]. It is possible that the high correlation between size and fitness in early stages of the experiment are caused by the mechanism described above. This does not mean that

other factors—such as enhanced nutrient import [57, 58]—do not improve the fitness, but rather that reduction in envelope-related costs might be one of them.

Although intended to capture the physiology of heterotrophic bacteria, the developed model could be extended to bacteria with other types of metabolism, such as photoautotrophs. Unlike the heterotrophs, the phototrophs do not directly consume the reduced carbon (sugar) but must instead produce it via fixation of the oxidized carbon source (CO_2). The model could accommodate this in two ways. One approach is to introduce an additional proteome sector that makes the precursor P_b , which will subsequently be used for the building block synthesis. That is, such bacterium consumes CO_2 and convert it to a precursor sugar, which is, in turn, transformed into the building block that makes the biomass. Next, the growth laws could be re-derived to infer the necessary model parameters.

Based on the preceding analysis, we propose the following scenario for the origins of the scaling of bacterial size and growth. Bacteria inhabit diverse environments that favor various cell shapes and sizes, due to size-selective predation, swimming efficiency in viscous environments, maximization of nutrient uptake rate, enhancing dispersal, and so on [41, 59]. The shape and size, in turn, set the maximal growth rate that a bacterium can achieve via investment trade-off between biosynthetic and envelope-producing processes. Alternative explanations are now discussed.

4.1. Gene-repertoire hypothesis

It was proposed that the larger bacteria have larger genomes and a greater gene repertoire, which allows them to metabolize a more diverse set of nutrients (or use them more efficiently) compared to their smaller counterparts [3]. There are a few problems with this idea. First, it is difficult to test this hypothesis because it does not offer a quantitative prediction on how genome length or gene number scaling should translate to the scaling of the growth rate. For example, should one expect the growth rate to scale with an identical exponent as gene number? If so, data in [3] refute this hypothesis, given that the growth rate scales with the power of 0.73 and gene number and genome size with the power of 0.35. On the other hand, if the growth rate scales with the square of the gene number, then their explanation seems more plausible. Second, even if the gene-repertoire hypothesis is true, it begs the question as to what causes genome size (or gene number) to scale with the power of 0.35 with cell mass instead of some other value. Third, while this hypothesis might in principle explain interspecific growth scaling, it cannot explain scaling of growth in experimental evolution as all data points correspond to the same species with the identical gene repertoire.

4.2. Internal diffusion constraints hypothesis

An alternative hypothesis is that larger cells grow faster because their cytoplasm is less crowded, thus alleviating constraints from the internal diffusion of macromolecules [45]. According to this hypothesis, growth is proportional to the abundance of metabolic proteins, implying that faster growth can be only achieved by increasing their abundance. However, increased abundance of effectors can slow down intracellular diffusion and thus reduce the encounter rates between cellular components. To offset this problem, an increase in the total mass of these effectors has to be accompanied by a faster increase of volume, such that the density of the cell decreases. On the contrary, volume and mass scale proportionally in our data set (slope of 0.93 ± 0.084 with adjusted $R^2 = 0.88$ is not significantly different from unity, $p = 0.412$), meaning that larger cells are not less dense. Furthermore, if selection optimizes the density of cytoplasm such that the rates of cellular reactions are maximized [60, 61], it is not clear why a decrease in density would not increase the mean time required for two proteins to collide with one another and, thus, decrease the rate of cellular reactions.

4.3. A link between the SMK law and cross-species growth rate scaling?

Finally, it is widely known that faster-growing cells tend to have larger volumes (colloquially referred to as the SMK or Schaechter–Maaloe–Kjeldgaard law), e.g. when growth is altered by rearing bacteria in media of differing nutrient quality [62], and one might argue that the scaling of growth rate across bacteria and across time in lines selected for fast growth is driven by the same underlying mechanism. The correlation between size and growth within species is explained by the threshold initiator model, which assumes that the cell divides once a critical threshold of a particular division protein is reached [38]. According to this model, the cell has to balance between synthesis of division protein and ribosomes. When the cell is reared in nutrient-rich conditions, it allocates a greater proteome mass fraction to ribosomal proteins to support fast growth, and fewer resources are left for the division protein. Therefore, the cell produces division protein more slowly than it grows in size, causing the average cell to be larger [63].

It is possible to co-opt this model in an attempt to explain cross-species correlation between size and growth. Here, the idea is that fast-growing species have a more efficient metabolism than their slow-growing counterparts and thus allocate more resources toward ribosomes, leading to the reduction in the rate of synthesis of the division protein. There are two problems with this explanation. First, it is unclear why selection would make metabolism of some bacteria efficient but not others. Perhaps this might be attributed to differences in environments

that they inhabit. For example, bacteria living in other organisms might not be selected for fast growth as a mechanism for preventing harm to the host, whereas selection in free-living species relentlessly promotes faster growth. We found that this is not so, and that the negative correlation between S/V and the growth rate holds both among host-associated bacteria as well as among the free-living species (section S1.12)). To illustrate this point, consider *Streptococcus pyogenes*, inhabiting throat and dividing in 25 min, and *Treponema denticola*, inhabiting mouth and dividing in 20 h. One is left wondering why selection in two species living in similar environments yields dissimilar metabolic efficiencies. Second, the SMK-based theory does not explain the difference in exponents between cross-species and within-species scaling. That is, the within-species scaling exponent of growth rate with S/V is smaller than -1 , contrary to the cross-species scaling.

Perhaps one could argue that the observed cross-species scaling is caused by the fact that small- S/V species require richer media, so that the perceived higher growth rate is merely the consequence of correlation between nutrient concentration and S/V . However, most of the maximal growth rates in our data were measured in media of similar nutrient quality containing yeast extract and peptone. Furthermore, many of the large- S/V species—like spirochetes and mollicutes—require rich media with bovine serum albumin.

One may also use the SMK-based model to explain correlated changes in cell size with the growth rate in lines experimentally evolved for faster growth. Cells start with fairly inefficient metabolism, and over the course of experimental evolution, their metabolic capabilities are gradually improved. Over time, this causes an increasing re-allocation of the proteome from metabolic proteins and division protein to ribosomal proteins, thus inflating the cell volume. This hypothesis predicts that if a faster-growing evolved line is forced to divide more slowly such that it has the same growth rate as the ancestor, it would also have the same cell size as the ancestor. However, rearing an evolved population in a chemostat at lower dilution rate still yields larger cells relative to the ancestor [64], implying that the increased cell size cannot be explained only as a by-product of the physiological response to growth rate change.

4.4. Constraints on growth imposed by molecular stochasticity?

One can speculate about alternative explanations for the growth scaling pattern that have not been entertained in the literature. Two such scenarios come to mind. First, small-celled bacterial species have a smaller number of proteins in the cell, meaning that the noise in gene expression and partitioning of molecules at cell division time is exacerbated. This stochasticity can disbalance the optimal

stoichiometry of biosynthetic pathways and, in turn, cause a reduction in the growth rate. While various theoretical studies have attempted to explain what aspect of the cell's physiology is the biggest contributor to the noise in growth rate and how the noise varies across growth conditions (e.g. [65, 66], we are not aware of an attempt to quantify how far below the deterministic limit is the growth rate depressed due to noise in gene expression and partitioning of molecules. Second, following the reasoning proposed in [67], fast-growing cells may be bigger because this reduces the noise in gene expression of metabolic proteins. The idea is that external ecological factors (such as availability of nutrients) determine the cell's growth rate independent of the cell size. Faster-growing cells have a greater mass fraction allocated to ribosomal proteins, implying that a smaller mass fraction is allocated to other metabolic proteins. If the cell volume does not change, the faster-growing cell will experience a greater noise in the expression of metabolic proteins, owing to the low-copy number of this sector. Hence, fast growers might have evolved larger sizes to dampen this stochasticity.

A problem with both of these hypotheses is that one would intuitively expect that the growth rate scales inversely with the volume of the cell. For example, if one imagines a cell as a bag of ribosomes that make themselves at rate κ and divides once critical threshold X is reached—assuming the predominant source of noise is random partitioning of molecules into the daughter cells at the cell division time [66], the expected cell division time is $\text{Log}[2]/\kappa + 1/(2\kappa X)$ (see section S1.16). If the number of molecules is proportional to the cell volume, one sees that increase in cell volume will decrease the cell division time until a plateau of deterministic cell division time $\text{Log}[2]/\kappa$ is reached. In contrast, judging from the preceding analysis, surface-to-volume seems to be a stronger predictor of the growth rate than the volume itself.

4.5. Relationship to previous studies

The theory developed here conflicts with three previously published conclusions. First, contrary to the assumption that the selection coefficient (i.e. the evolutionary cost) of a particular cellular feature is directly proportional to the fraction of total resources invested in the production of the trait [4, 68], here we show that this is not necessarily true, and that it can also depend on the rates of cellular reactions (equations (S1.2)–(S1.4)). This is because the costs of the trait also have to include the costs of machinery that builds the trait (in this case, the envelope-producing enzyme), and the amount of enzymes produced depends on how fast it operates relative to other proteome components: If the enzyme is slow, then a large fraction of the proteome has to be allocated to it to achieve an appropriate flux. Second,

although a recently published analysis [69] concluded that there is no correlation between cell diameter and growth rate across prokaryotes, our analysis reveals that growth rate correlates with S/V , at least when one focuses on bacteria for which both linear dimensions are available.

Third, it was argued that the observed scaling of various bacterial features necessarily imposes the upper limit on how large a bacterium can be [70]. According to this view, the ribosomes have to replicate themselves and all other proteins within the cell doubling time. So, if ribosomes, other proteins, and doubling time scale with different exponents, it is possible for a cell to reach a particular volume at which it does not have enough ribosomes to replicate the entire proteome within the doubling time inferred from a specific power function—an event deemed the ‘ribosome catastrophe’. However, given that many functions can fit the same data, it is unclear whether fitting pure power functions is adequate. Indeed, our model can explain the scaling of both proteome and growth rate without any fundamental cell size limit. Of course, there might be many reasons why bacteria cannot evolve extreme cell sizes, but inferring this limit from the scaling laws within extant bacterial species may be a questionable approach.

Lastly, although it may be surprising that the growth rate does not increase with the cell volume as suggested in previous studies [5], the absence of a relationship is caused by the fact that our dataset contains spirochetes and helical mollicutes which were absent in previous studies. These organisms have an average bacteria-like volumes, but rather large surface-to-volume ratio because they are shaped as thin and long coils. Thus, their slow growth rates and fairly ‘normal’ volumes obscure the purported trade-off. However, upon exclusion of helical organisms, one retrieves a similar scaling relationship between the growth rate and the cell volume as reported in [5].

4.6. Consequences for the scaling of genomic features with cell size

Our model also offers a causal explanation for the correlation between genomic features associated with translation and growth rate. For example, faster-growing bacteria tend to have a greater number of rRNA operons [71]. According to our theory, bacteria grow slowly because they cannot allocate enough resources to ribosomes, given that other surface-related constraints have to be satisfied. Suppose the high copy number of rRNA genes is caused by the need to meet the high demand for ribosomes. In that case, species with large S/V will have no selective advantage in possessing additional gene copies, meaning that they will be molded by selection to reduce the costs of replicating the added DNA. The same explanation holds true for tRNA genes. Unfortunately, our theory does not yield the precise expectation for the scaling of genomic features

because our model does not include the genome as a resource component, so this hypothesis cannot be tested rigorously at the moment. Nonetheless, we find a negative log–log scaling between S/V of the species and its rRNA ($p < 10^{-9}$, adjusted $R^2 = 0.18$) and tRNA ($p < 10^{-13}$, adjusted $R^2 = 0.26$) gene number (see section S1.13 in S1 appendix for details).

4.7. On the universality of the scaling of growth rate

The model presented here is not universal, as it cannot account for the scaling of growth across the entire Tree of Life, most notably eukaryotes. We would expect the growth rate to monotonically increase as S/V decreases, but we know that this is not true because eukaryotes with much smaller S/V have growth rates that decrease with size [5, 72]. Even large bacteria, like *Metabacterium polyspora* with a volume of $480 \mu\text{m}^3$, have doubling times measured in days [73]. Similarly, although our theory can explain size changes in *E. coli* lines selected for fast growth, it fails to account for size changes in eukaryotes under the same selective pressure. For example, propagation of *Kluyveromyces marxianus* in pH-auxostat leads to an increase in S/V , which contradicts our theory [74]. It appears that, as the cell size increases, a point is reached when other constraints become more relevant in setting the growth rate. While it is not known where such a tipping point occurs, what causes it, and how the growth rate becomes limited in its wake, one can entertain a few possibilities.

First, the membrane may become saturated with nutrient uptake sites (i.e. transporters), and any further increase in cell size will cause the rate of nutrient supply to increase in proportion to the surface area of the cell (scaling with $V^{2/3}$). But given that the demand for nutrients is proportional to the cell volume (scaling with V), larger cells will become increasingly deficient in material needed to build the biomass, thus causing their growth rate to decrease with the cell size (scaling with $V^{2/3}/V$ or $V^{-1/3}$) (chapter 3 of [23]). Second, if the rate at which the cell consumes nutrients from its vicinity is faster than the rate at which nutrients diffuse toward the cell, the growth of the cell becomes diffusion-limited. The flux of nutrients toward the cell—and so the uptake rate—scales with the linear dimension of the cell ($V^{1/3}$) whereas the demand for nutrients scales with V , implying that the growth rate ought to decrease with cell size and scale with $V^{-2/3}$ [54]. Third, once inside the cell, the nutrients have to be distributed throughout the cell’s interior. Because the rate at which diffusion equilibrates the concentration of nutrients in the cell decreases with the size of the cell [75], the rates of biochemical reactions that consume the nutrient and grow the cell ought to be slow in large organisms. Fourth, in photoautotrophic organisms, the absorptive capability of pigments packed in a cell is always lower than that of the equal concentration

of pigments in a solution, and this ‘package effect’ becomes exaggerated as the cell size increases. Thus, large-celled organisms will have lower efficiency of photosynthesis and consequently have lower growth rates [10].

The large cells can evolve a variety of adaptations that alleviate the physical challenges associated with being large. The saturation of membrane with nutrient uptake sites can be resolved by evolving membrane invaginations that increase the surface area without affecting the volume or freeing-up membrane space by internalizing part of the membrane-bound metabolism [76, 77]; The nutrient-depleted region around the cell can be replenished by swimming because the nutrient influx is now the result of both diffusion and advection [78]; the slow intracellular transport can be sped up by cytoplasmic streaming, whereby the movement of vesicles along the cytoskeletal elements drags the cytoplasm and increases the rate of mixing [79]; the self-shading of pigment molecules can be reduced by changing the shape of the cell [80]. Nonetheless, the mitigation of physical limits always comes with the cost: intracellular vesicles require the synthesis of additional lipids, swimming demands costly flagella, while enhanced mixing via active transport depends on elaborate cytoskeleton and molecular motors. Hence, the concomitant costs of these adaptations conspire to depress growth rate below the value that would have been obtained in the absence of physical constraint. Moreover, even if one sets aside the costs, the adaptations may impose additional constraints. For instance, the intracellular diffusion limit may be circumvented by establishing an active-transport system, but this also imposes other transport-related constraints on the rate of growth [9]. Given that the scaling exponents tend to vary across the Tree of Life [5], the scaling patterns are likely affected by multiple constraints acting in unison. Advances in integrating multiple physical constraints acting at once have been made [81], but the theory that mechanistically unifies them and derives expected scaling laws remains to be developed.

5. Conclusion

It is widely known that smaller eukaryotes tend to grow faster than larger ones. However, this trend does not hold in bacteria, where small-celled species grow slower. We find that small bacteria compared to their larger counterparts have to invest a greater fraction of the total resources to the cell envelope owing to their large surface-to-volume ratio, leaving fewer resources for internal biosynthetic processes that build the cell. These predictions are corroborated by comparison with growth rate data across more than 200 bacterial species.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://github.com/BogiTrick/growth_scaling_envelope.

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Conflict of interest

The authors declare that they have no competing interests.

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